



From event logs to goals: a systematic literature review of goal-oriented process mining

Mahdi Ghasemi¹ · Daniel Amyot¹

Received: 2 March 2017 / Accepted: 27 December 2018 / Published online: 7 January 2019
© Springer-Verlag London Ltd., part of Springer Nature 2019

Abstract

Process mining helps infer valuable insights about business processes using event logs, whereas goal modeling focuses on the representation and analysis of competing goals of stakeholders and systems. Although there are clear benefits in mining the goals of existing processes, goal-oriented approaches that consider logs during model construction are still rare. Process mining techniques, when generalizing large instance-level data into process models, can be considered as a data-driven complement to use case/scenario elicitation. Requirements engineers can exploit process mining techniques to find new system or process requirements in order to align current practices and desired ones. This paper provides a systemic literature review, based on 24 papers rigorously selected from four popular search engines in 2018, to assess the state of goal-oriented process mining. Through two research questions, the review highlights that the use of process mining in association with goals does not yet have a coherent line of research, whereas intention mining (where goal models are mined) shows a meaningful trace of research. Research about performance indicators measuring goals associated with process mining is also sparse. Although the number of publications in process mining and goal modeling is trending up, goal mining and goal-oriented process mining remain modest research areas. Yet, synergetic effects achievable by combining goals and process mining can potentially augment the precision, rationality and interpretability of mined models and eventually improve opportunities to satisfy system stakeholders.

Keywords Business process management · Event logs · Goal mining · Goal modeling · Intention mining · Performance indicators · Process mining · Requirements engineering · Systematic literature review

1 Introduction

In organizations, *processes* are performed to meet the requirements of all corresponding stakeholders and to achieve predefined *goals*. Processes are composed of activities that often can be logged in files or databases during their execution. Process logging becomes increasingly popular for enabling process optimization and compliance. Yet, finding some efficient ways of monitoring processes, while considering the (often conflicting) goals of all stakeholders, is an intricate task.

On the one hand, *process mining* is an emerging research area and an evidence-based approach looking to exploit event logs and to infer valuable insights about processes. In particular, process mining plays an important role as a bridge between traditional model-based process analysis (e.g., simulation) and data analysis techniques (e.g., data mining). This causes a huge demand for data scientists who are not only able to analyze big data, but also to relate them to actual operational processes. According to van der Aalst [100], process mining techniques fall into one of the three categories: (1) *discovery*, where a model is produced using the event logs; (2) *conformance*, where the data generated from the model are compared with the actual data in event logs to assess reality against the model; and (3) *enhancement*, where an existing process model is improved and/or extended.

On the other hand, *goal modeling* is a requirements engineering approach used to support heuristic, qualitative or formal reasoning schemes [46]. A goal is an objective

✉ Daniel Amyot
damyot@uottawa.ca
Mahdi Ghasemi
mghas020@uottawa.ca

¹ School of EECS, University of Ottawa, 800 King Edward, Ottawa, ON K1N 6N5, Canada

meant to be achieved by the system under consideration or one of its stakeholders. Goals are typically modeled by some fundamental features such as their type and attributes, and by their *links* to other goals and to other elements of a requirements model [106]. During the last decade, several methods have been proposed that prescribe the application of goal-oriented requirements engineering techniques for supporting business process management activities, in particular business process modeling [67, 80]. While the process-oriented modeling languages primarily focus on “how,” “what,” “where,” “who” and especially “when” questions, goal-oriented modeling focuses on answering “who,” “what” and especially “why” questions, hence offering a way to document intentions and rationales. A goal-oriented approach leads the modelers to consider the opportunities that stakeholders look for and vulnerabilities that they try to avoid, whereas modeling and answering “what” questions help identify capabilities, services and architectures required to satisfy stakeholder goals [6].

Process mining constitutes a form of requirements engineering, from requirements elicitation to validation. In particular, process discovery uses a form of crowdsourcing (where logs are created from many system users performing tasks) and helps generate as-is process models that reflect real behaviors recorded in logs. Process mining can be used as a data-driven substitute or complement to use case/scenario elicitation. In scenario-based requirements engineering [96], scenarios represent paths of possible behaviors through a use case, and they are investigated to elaborate requirements. Scenario-based requirements engineering and process mining both consider the question of how instance-level data can be generalized into models, except that in process mining such data are orders of magnitude larger. Scenario-based techniques (e.g., [91, 97, 98]), however, do not scale well to process mining-size problems.

Scenarios may be used to validate requirements, as ‘test data’ collected from observable practice, against which the operation of a new system can be checked [97]. Conformance checking is, also, a means to expose where the real process has deviated from a desired or prescribed model. Process conformance and enhancement enable requirements engineers to find new system/process requirements that will improve the alignment between current practices and desired ones. When a deviation in an event log is considerably frequent, this situation should trigger the requirements engineer to re-assess the suitability of the system’s goals and processes. The reasons behind frequent deviations could lead to new or modified goals and requirements. This is where *goal mining* can contribute to the work of requirements engineers, by supporting the discovery of goals from execution logs and improving the soundness of goal models, requirements and processes.

One limitation of using execution logs to reflect user behavior and as-is processes is that the processes should be automated and instrumented precisely. However, when they exist, such logs might be considered as a complement (or even substitute) to observations and interviews with process participants.

One important research trend, aligned with increasing research interest on big data, is the elicitation of requirements from massive collections of data [95]. Recent work on data-driven and user-centered approaches in software requirements engineering [51, 69] is part of this trend. These approaches [51, 69] focus on the identification, prioritization and management of software requirements based mostly on online feedback and reviews about software products that users can easily submit in app stores, social media or user groups. The current review is, however, different from such approaches as it focuses on execution logs, processes and goals.

Combining goals with processes leads to many known benefits. For example, Horkoff et al. [48] recently reviewed 243 papers to assess how goal modeling can impact downstream activities such as architectural decisions and process/system development. Many identified relationships suggest that the use of goal models can have a major positive influence on:

- the content and structure of processes, based on business and trade-off analysis;
- process quality, through measurement and compliance;
- decision making, business intelligence and predictions;
- process alignment, configuration, adaptation and evolution.

Process alignment with goals can also be checked to some extent at the structural level with the help of consistency rules [2].

Relevant to the core of our work, Papadimitriou et al. [76] conducted a review to provide a common reference point to the main concepts and challenges that should be defined when someone intends to approach a data analysis task from a goal-oriented point of view. To this end, that review covers goal-aware systems as a whole and does not focus on a specific area or discipline. Goal-aware systems are able to perform three main tasks, all of which are related to goals: (1) *goal model construction*, to model and store goals, (2) *goal inference*, to identify the goals given a set of observed actions performed by users and (3) *goal exploitation*, to use the inferred goals for adapting the system’s functionality according to the goals. In other words, a goal-aware system requires the mechanisms to record the goal-related data, analyze it and then use the analysis results to recognize the actors’ goal, and respond accordingly [76]. That review also establishes a common

terminology and provides formal definitions for a number of concepts such as *environment*, *action*, *goal* (soft/hard) and *plan*. Then, employing those definitions, the most common techniques for constructing models, inferring the goals and exploiting the results are presented.

In contrast to Papadimitriou et al. [76], who covered goal-aware systems at a high level (with an emphasis on information retrieval) without getting into the details of specific techniques, our review focuses on one type of goal inference approach, not covered in their paper. Our review aims to assess existing work and opportunities where processes and their goals are mined from evidence such as process execution logs. Considering that process mining is the main field of research that focuses specifically on inferring knowledge from event logs, we study the works related to goal-oriented process mining. Such assessment, which has not been done in the past, requires an understanding of research that combines goal-oriented concepts with process mining techniques. In such a context, a systematic literature review (SLR) concerned with the intersection of both areas, can help provide insights into the above objective. An SLR is a methodologically rigorous review of relevant research results [61]. In such a review, systematic, explicit and trackable methods are employed to search, identify, select, appraise and sensitize relevant research activities in order to answer some clearly formulated research questions. This paper's review is systematic, focuses on the research where process mining and goal modeling are combined and provides accurate insight about the status of goal-oriented process mining activities. Such review is beneficial to requirements engineers interested in obtaining evidence-based definitions of existing processes and their goals prior to defining requirements for their modification or improvement.

The organization of this paper is as follows. The next section presents a brief overview of process mining and goal-oriented modeling. In Sect. 3, existing papers related to goal-oriented process mining are systematically collected according to the needs of two research questions. This part highlights the absence of existing review related to goal-oriented process mining, as well as the relative paucity of work done at the intersection of process mining and goal-oriented modeling. Section 4 analyses the contributions of 24 relevant papers along three categories: goal modeling, intention mining and performance indicators. Section 5 synthesizes and reflects on the selected papers and presents the value added by this review. In addition, Sect. 6 discusses several threats to the validity of this review. Finally, Sect. 7 provides conclusions, including a discussion of the impact on research and practice, and highlights important future work items for the requirements engineering community.

2 Overview of process mining and of goal-oriented modeling

This literature review aims to take into consideration the papers focusing on goal modeling and process mining at the same time. First, an overview of these individual research areas will be helpful to highlight the main concepts of each of these domains.

2.1 Process mining

Cook and Wolf [19] published the first article on process mining while they were working on discovering process models from event logs in the context of software engineering [3]. Since then, several groups focused on process mining models and developed different algorithms and implemented tools and frameworks. As Ghasemi and Amyot [39] demonstrated in a recent literature review, research in process mining has gained much interest over the past decade with 466 unique papers published in 2015, up from 294 papers in 2014.

Process mining is providing some new techniques to infer valuable knowledge and insights from event logs. For example, the electronic patient records in a healthcare system or the transaction logs of an enterprise resource planning system can be used to discover models describing processes, organizations and products [100]. Moreover, such event logs can also be used to compare reality with prior models to determine their mutual alignment [89].

In general, process mining activities are distinguished into three main types: (1) discovery, (2) conformance and (3) enhancement (see Fig. 1). In the *discovery* type, the aim is to produce a process model using event logs. There is no prior model involved in a process discovery technique.

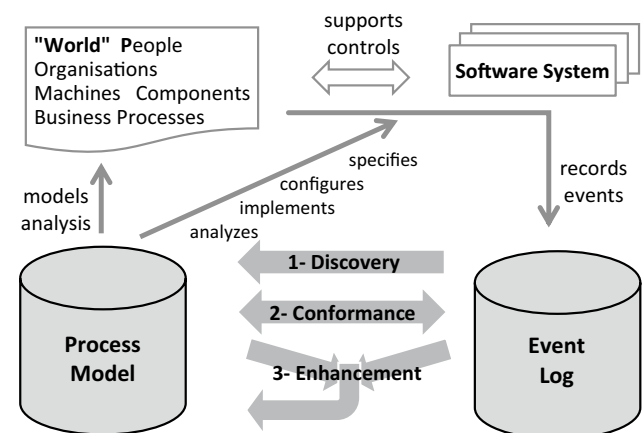


Fig. 1 Three types of process mining: (1) discovery, (2) conformance and (3) enhancement Adapted from [100]

The inferred model should be able to describe the observed behavior of a process. In *conformance*, an existing process model is compared with observed behavior in the event logs. If there is a deviation between the model and the event log, it can be further analyzed, for example, to detect activities in the model that do not exist in the event log or vice versa [65, 89]. In *enhancement*, the target is to change or improve an a priori model. Note that it is supposed that a model, either produced manually or discovered, already exists [100, 101].

There are many process mining algorithms proposed by researchers and practitioners. Each of these algorithms has different performances for different kinds of processes. As a result, it is not easy to select an appropriate process mining algorithm for a given application domain [111]. The α -algorithm [104] shows many of the general ideas used by many process mining algorithms and aids to understand the concept of process discovery. However, the α -algorithm is not a very practical mining technique as it has some problems with noise, infrequent behavior, incomplete behavior and complex routing constructs [100]. Heuristic mining algorithms [112] use a representation like graphs whose nodes capture activities and arcs that describe causal dependencies. These algorithms consider frequencies of events and sequences when making a process model.

The *ProM* framework [83, 105], an extensible environment, is the most popular tool for process mining. *ProM* is flexible in terms of the input and output formats and allows reusing code throughout the implementation of novel process mining ideas. Process mining capabilities are also provided by a growing number of commercial analysis tools such as *Disco* [33].

2.2 Goal modeling

Goals are objectives that stakeholders and systems aim to achieve. Goals can be used at different levels of abstraction. Goal-oriented requirements engineering is concerned with the use of goals for requirements elicitation, elaboration, structuring, specification, analysis, negotiation, documentation and modification [5, 106]. Whereas processes excel at answering “when” questions, goals answer “why” questions that are not handled by process models, and hence both views are complementary.

A *goal model* is a directed graph that shows how goals contribute to each other [16]. AND refinement links relate a goal to decomposing sub-goals. Here the parent goal is satisfied if and only if all sub-goals are satisfied. OR refinement links relate a goal to some alternative refinements. This means that satisfying one of the sub-goals is sufficient for satisfying the parent goal. Contribution links between goals capture knowledge about the extent to which goals positively or negatively influence other goals. In this context, a conflict between two goals is present

when the satisfaction of one goal may prevent the satisfaction of the other [106]. In case a goal contributes negatively to another goal in the goal model and the former one is satisfied, then the latter is unsatisfied. These kinds of rules are used for qualitative reasoning about goal satisfaction, e.g., in the Goal-oriented Requirement Language (GRL) [5].

In the context of requirements engineering, one benefit of goal modeling is to support heuristic, qualitative or formal reasoning schemes. Moreover, one can verify that the requirements entail predefined goals [106]. Goals and requirements are elicited from multiple stakeholders with different viewpoints. Therefore, a wide range of conflicts can arise during analysis [109]. In order to reason about conflicting goals, one of the major benefits of goal modeling is to support trade-off analysis, targeting a solution that brings an appropriate balance when not all goals can be satisfied simultaneously. Goal models are reputed to be efficient in identifying and resolving conflicting concerns [109], and in producing robust requirements through obstacle analysis [107, 110].

The goals’ *intrinsic features* such as their types and attributes, together with their *links* to other goals and to other elements (e.g., containing actors, indicators and requirements), make a goal model [106].

Several classification axes and types of goals have been proposed in the literature. *Functional* goals underlie services that should be delivered by the system. The functional goals characterize the expected system’s behaviors, for example “Issue a receipt for every transaction.” In contrast, *non-functional* goals talk about expected qualities of the system, for example “the receipt should be easy to read.” *Non-functional* goals such as security, safety, usability, flexibility, customizability, and so forth, are often captured with *soft goals* [6].

Several languages and notations have been developed to explicitly model goals and their relationships by the requirement engineering community. For example, popular languages include GRL [4], Keep All Objects Satisfied (KAOS) [108], MAP [87] the NFR framework [18], *i** [116], the business intelligence model (BIM) [47] and tropos [14]. In 2008, GRL was recognized as an international standard for goal-oriented modeling, as part of a recommendation of the International Telecommunication Union named User Requirements Notation (URN) [50]. It is noteworthy that URN is a standard notation capable of dealing with activity process models (in the form of Use Case Maps) and with goal models. URN also supports key performance indicators (KPIs) and other concepts to measure and align processes and goals [82].

Another goal language, MAP, was proposed by Rolland et al. [87] to address a new category of intentional process models in the context of business requirements engineering. As MAP is the notation used in *intention mining* research

activities (to be discussed in Sect. 4.2), it is important to explain that language here.

A process, while performing, is not limited to linear activities; actors, based on their context, have a variety of choices for performing a task. MAP models direct the actor by proposing dynamic choices aligned with their intentions. A MAP process model is a labeled directed graph that consists of *intentions* as nodes and *strategies* as arcs between intentions. The directed nature of the graph shows how the intentions can follow one another. Figure 2 shows a simple example of a MAP intentional model for buying a smart phone.

There are three key elements that make a MAP: *Intention*, *Strategy* and *Section*. An *Intention* is a goal that can be achieved by performing some activities. For example, there are two intermediate intentions in Fig. 2: selection of a smart phone, and buying the smart phone. A *Strategy* is a means to achieve an intention. For example, “By asking friends” is a strategy to achieve the phone selection intention. Finally, a *Section* is a triplet composed of a *Source Intention*, a *Target Intention* and a *Strategy* linking them. Therefore, a MAP consists of a several sections, each of which being represented as a triplet $\langle I_s, I_t, S_{st} \rangle$. In our example, one such section is $\langle \text{selection of a smart phone, buying the smart phone and by online shopping} \rangle$.

In a MAP, there might be several alternative links from I_s to I_t , each corresponding to an explicit strategy (multi-thread flow) to reach I_t from I_s . There might also be several strategies from different intentions to reach an intention I_t (multi-flow paths to achieve an intention). Finally, a MAP can contain reflexive flows, i.e., I_s and I_t are the same intention. There are two special intentions called *Start* and *Stop*, which represent, respectively, the intentions to start crossing the map and to stop crossing it. As there can be several strategies that could be selected between two successive

intentions, there are usually many paths in the graph from *Start* to *Stop*.

Rolland and Salinesi [88] have introduced and discussed goal/strategy MAPs as an example of goal models conceived to meet the challenges faced by *multi-purpose* systems, i.e., systems that incorporate variability in the functionality they provide and that can self-adapt to different situations. Kraiem et al. [62] proposed a novel approach for mapping intentional process models represented in MAP to business process models expressed in the business process model and notation (BPMN) [73]. The proposed approach links the MAP elements to BPMN elements in order to align the intentional and operational levels.

3 Literature review of goal-oriented process mining

This section aims to review the papers related to goal-oriented process mining. The studies focusing on the combination of goal modeling and the capabilities of process mining techniques are discussed. The methodology of this SLR, including the research questions, the steps for selecting relevant work, the identification of search engines, the selection criteria and the inclusion/exclusion criteria, is provided. Then, among the selected papers, those that are related to the topic of this research are categorized and their contributions are summarized.

3.1 Research questions

Inspired by the work of Kitchenham et al. [61], this systematic literature review aims to answer the following research questions:

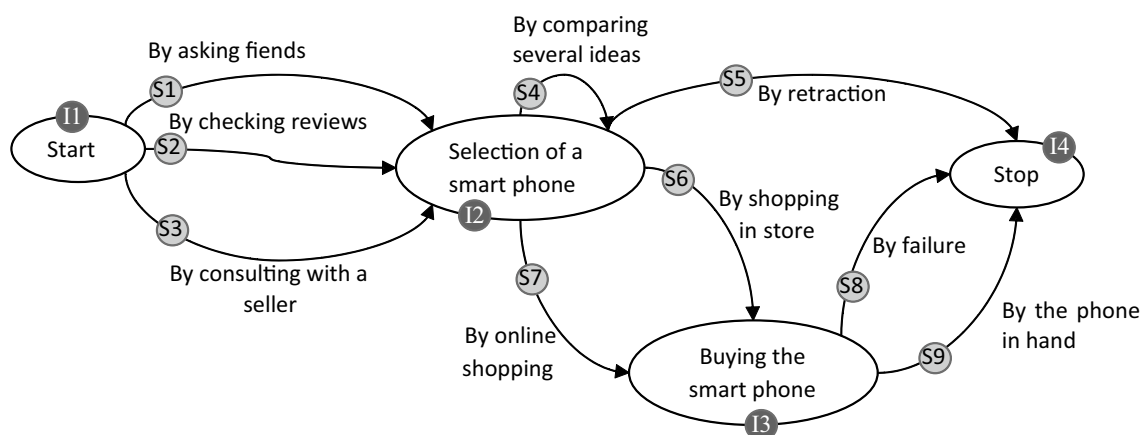


Fig. 2 A MAP intentional model for guiding users who want to buy smart phones. There are four intentions (I1–I4, including Start and Stop) and nine strategies (S1–S9) available to achieve the intentions

RQ1. What is the share of goal-oriented approaches in process mining studies? This question aims to determine whether goal-oriented process mining is a mature research field or whether there is currently a gap that exists.

RQ2. How are the studies conducted on the joint goal modeling and process mining techniques categorized, and what are the main contributions of the papers in these categories?

3.2 Identification of search engines

A recent literature review of process mining reviews [39] considered four of the most relevant search engines in information technologies, namely Scopus, Google Scholar, IEEE Explore and Academic Search Complete (ASC), together with PubMed, Embase, Clinical Evidence and the Cochrane Library, which are four important healthcare search engines. That review showed that 73.5% and 42.6% of the unique papers related to process mining within these eight search engines could be returned by Scopus and Google Scholar, respectively. In addition, 95.6% are covered by at least one of these two engines and 19.7% of the papers are mutually found by both. Therefore, if one simply uses these two search engines, he/she will miss only about 5% of the papers, which were returned only by other six search engines (mostly IEEE explore and ASC). However, to be on the safe side, in addition to Scopus and Google Scholar, IEEE Explore and PubMed were used in this research. As this review targeted the papers positioned in the intersection of process mining and goal modeling areas, the desired papers are necessarily positioned in the process mining area.

3.3 Keywords and queries

As the authors were looking for papers focusing on process mining and goal modeling, they defined two sets of keywords, i.e., one for each area. Then, the following queries were defined in all four search engines looking for papers that include at least one of the keywords from each set:

```
(goal[Keywords] OR "goal-oriented" OR "Goal modelling" OR "Goal modeling" OR "Goal model" OR "Goal mining" OR "Goal monitoring" OR "requirements engineering" OR "user requirements notation" OR KAOS OR istar OR "i-star" OR "i star" OR "NFR Framework" OR Tropos OR GRL OR intention OR intentional OR "performance indicator" OR KPI)
```

AND

```
("Process mining" OR "Process discovery" OR "conformance checking" OR "event log")
```

Table 1 Number of papers returned by the search engines

	Search within	Google scholar	Scopus	IEEE Explore	PubMed
1	Limited to title/abstract/keywords	11	81	10	7
2	Duplicates removed, across all engines	92			

Note that this query was constructed iteratively, i.e., the authors started with some essential keywords such as goal modeling, and then the returned papers informed them of other relevant and related keywords such as intention and requirements engineering.

Initially, the query within the full text of the papers was run. This query returned thousands of papers, but most of them were irrelevant. In order to end up with manageable, meaningful and relevant results, the search was limited to metadata (title, abstract and keywords). As searching within abstracts and keywords is not supported by Google Scholar, queries on that engine were limited to only the title (patents and citations were also excluded). Also, given that the word “goal” is a popular word used in almost all paper abstracts (e.g., “*the goal of this paper is...*”), the queries were limited to look for this specific word only within the keywords (the other search terms were searched for title, abstract and keywords). This constraint reduced the number of results dramatically and enabled the authors to analyze the related papers while avoiding noise from less relevant papers. To conserve the ability of finding as many papers as possible, the review was not consciously limited to a particular time period.

The results of the query are shown in the first row of Table 1. Finally, after removing duplicates among the results of the four search engines, 92 papers remained.

3.4 Frequency analysis of publications

In order to find the position of *goal-oriented modeling* in the process mining research area (and vice versa), two more queries were run with Scopus looking for the aforementioned sets of keywords separately. This time, the keywords were {"GOAL-ORIENTED" OR "GOAL MODEL" OR "INTENTION MINING" OR "USER REQUIREMENTS NOTATION" OR KAOS OR ISTAR OR "I-STAR" OR "I STAR" OR "NFR FRAMEWORK" OR TROPOS OR GRL OR "GOAL MINING" OR "GOAL MONITORING" OR "INTENTIONAL MODELING" OR "GOAL MODELING" OR "GOAL MODELING"} (this time only), within the indexed keywords of journal and conference papers. KPI and "PERFORMANCE INDICATOR" were omitted because these keywords independently do not refer to the goal modeling field. Also, the same engine was queried run with the keywords {"PROCESS MINING" OR "PROCESS DISCOVERY" OR "CONFORMANCE CHECKING" OR "EVENT LOG"}. The numbers of papers returned through each query were 3048 and 2203, respectively. These high numbers clearly show that the papers at the intersection of these areas of research (92) are quite sparse.

In addition, the authors considered the frequencies of all keywords that have been indexed by the 2203 papers related to process mining returned by Scopus. It is not surprising that the keywords "*process mining*" and "*data mining*" are the top ones, by far. However, the word "goal" did not show up within the top 160 keywords that were ranked based on frequencies.

3.5 Selection criteria and related papers

In order to exclude the papers that are irrelevant to our two research questions, exclusion criteria were defined.

These criteria helped exclude papers that:

- only cited process mining and goal modeling or their derivatives to acknowledge their existence or to mention them in passing in some examples; or
- were related to only one of the two areas considered; or
- were neither clearly related to process mining nor to goal modeling; or
- were not written in English.

The aforementioned criteria resulted in 24 papers (including one thesis) in which a research activity related to the topic of this review was reported (see Table 2). A quality assessment confirmed that the papers (especially those returned by Google Scholar) had sufficient scientific content and were not coming from predatory journals or conferences. In the following section, these specific papers will be reviewed.

4 Analysis of selected papers

After considering the selected papers and clustering their main subjects, the papers were categorized into three main areas: goal modeling and requirements elicitation, intention mining and key performance indicators (KPI). In this section, the selected papers are analyzed and their contributions are summarized.

4.1 Goal modeling and requirements elicitation

Business processes are designed to achieve certain goals, namely *process goals*. Based on these process goals, individual agents are assigned to perform the related tasks composing the process. However, the agents generally belong to different sections/units and have different interests, and their behavior is not usually observable. In the real world, agents have intentions reflecting their own interests, namely *agent goals*. In order to satisfy these goals, the agents may not always behave according to a designed process. The inconsistencies or, in some cases, conflicts between process goals and agent goals often result in a reduced efficiency or effectiveness of the executed process [115]. However, traditional business process management ignores the feature of self-interest in the business process and adopts an inter-organization workflow perspective [99]. This phenomenon is common in the context of agent-oriented business processes, which have been the focus of the literature in some research domains such as healthcare [1], supply chain management [9], financial fraud management [60] and others. Process mining could have a valuable contribution in addressing this common problem. However, in the literature, there is a lack of studies on how to discover agent goals in business processes. To fill this gap and to address that problem, Yan et al. [115] proposed an agent-oriented goal mining approach for modeling, discovering and analyzing agent goals in executed business processes using event logs. This is clearly positioned in the context of process discovery as a type of process mining activity.

Yan et al. [115] used the classical goal reasoning framework of *belief–desire–intention* (BDI) logic [35] to analyze an agent's choices using event logs with domain data from the business process. They have explained some supporting reasons to show that the BDI model has several advanced features that meet the requirements of an agent's goal mining, compared to other computational decision models. Yan et al. [115] have used the framework shown in Fig. 3 within a sample process used to invite external reviewers to evaluate loan requests.

Three activities are executed in the process. First, an agent assigns a reviewer for each loan request. Then, a reviewer performs the activity of evaluating the request, and finally

Table 2 Selected papers related to goal-oriented process mining, with citations from Google Scholar on December 21, 2018

Category	Paper	Code	Makes reference to	Year	Source	Cited	Goal model representation	Proposed goal mining algorithm?	Case study/evaluation?
Goal modeling and requirements elicitation	[8]	P1	–	2012	Conf: ICITCS	0	None	No	No
	[7]	P2	–	2012	Conf: ICCSA	0	None	Yes	Yes
	[115]	P3	–	2014	Journ: TMIS	3	AND/OR graph	Yes	Yes
	[45]	P4	–	2015	Conf: SEKE	0	AND/OR graph	Yes	Yes
	[81]	P5	P12	2015	Conf: BPM	3	KAOS	No	Yes
	[75]	P6	–	2016	Journ: SoSyM	9	None	No	Yes
	[11]	P7	–	2017	Conf: BPM	4	None	No	No
	[92]	P8	–	2017	Conf: ER	0	AND/OR graph	Yes	Yes
	[22]	P9	–	2017	Conf: RE	1	None	No	Case study
	[114]	P10	–	2017	Journ: J Healthc Eng	1	None	Yes	Yes
Intention mining	[52]	P11	P12	2013	Conf: RCIS	6	Mentioned and compared some of them		Yes
	[55]	P12	P13, P11	2013	Conf: RCIS	23	MAP	Yes	Yes
	[54]	P13	P11, P12	2013	Conf: EMMSAD	21	Mentioned and compared some of them		Yes
	[58]	P14	P11, P12, P13	2014	Conf: RCIS	3	MAP	Yes	Yes
	[24]	P15	P11, P12, P13	2014	Journ: IJISMD	7	MAP	Yes	Yes
	[57]	P16	P11, P12, P13	2014	Conf: MSR	17	MAP	Yes	Yes
	[56]	P17	P12, P16	2014	Conf: BPMDS	11	MAP	Compare	Yes
	[53]	P18	P11, P12, P13, P14, P16, P17	2014	Thesis: U. Paris-Est	2	MAP	Yes	Yes
	[29]	P19	P12	2014	Conf: CAiSE	6	MAP	Yes	Yes
	[59]	P20	P14, P15, P16, P17, P18	2015	Journ: IJISMD	3	MAP	Yes	Yes
KPI	[63]	P21	–	2013	Conf: CBI	9	Performance analysis		Yes
	[64]	P22	P21	2014	Conf: CBI	2	Performance analysis		Yes
	[23]	P23	–	2017	Conf: OTM	3	Model redesigning with KPIs		Yes
	[17]	P24	–	2017	Journ: DSS	2	Performance analysis		Yes

the request is either assigned to a new reviewer or sent to the payment agent. The authors have elaborated this example, which has been supported by some event logs used to mine the agent goals (number 3 in Fig. 3).

In their proposed goal mining algorithm, the goals are expressed as decision tree rules. In order to discover an agent's goal, Yan et al. [115] adopt an approach to discover the agent's desire-accessible worlds based on the agent's previous activities. They assume that if an agent has chosen an activity, she knows the effects of the activity and believes that she achieves her goal by performing this activity. Thus, one can learn the agent's goal by classifying the agent's activities in different situations. In other words, the agent's desire-accessible worlds can be identified by machine learning methods. Yan et al. [115] adopted the decision tree algorithm (C4.5) to construct the desire-accessible worlds, i.e., the agent's goal, within

their example. As inputs, their proposed algorithm needs event logs and the agent-oriented Petri net process model discovered by process mining algorithms. The outputs of the algorithm are the agents' goals at each state.

Yan et al. [115] have evaluated their framework by simulation techniques and compared their goal mining-based approach with the best-performer-based approach, using *t* test for testing the improvement hypothesis. Based on their case study, in terms of improving the effectiveness ($= 1 - \text{Number of requests wrongly graded} / \text{Number of requests}$), their goal mining-based approach has not significantly improved effectiveness, comparing with the best-performer-based approach. However, their approach has significantly improved efficiency ($= \text{Number of requests} / \text{Evaluation times}$).

Along with the goal-oriented approach of Yan et al. [115], Armentano and Amandi [7] focused on the automatic

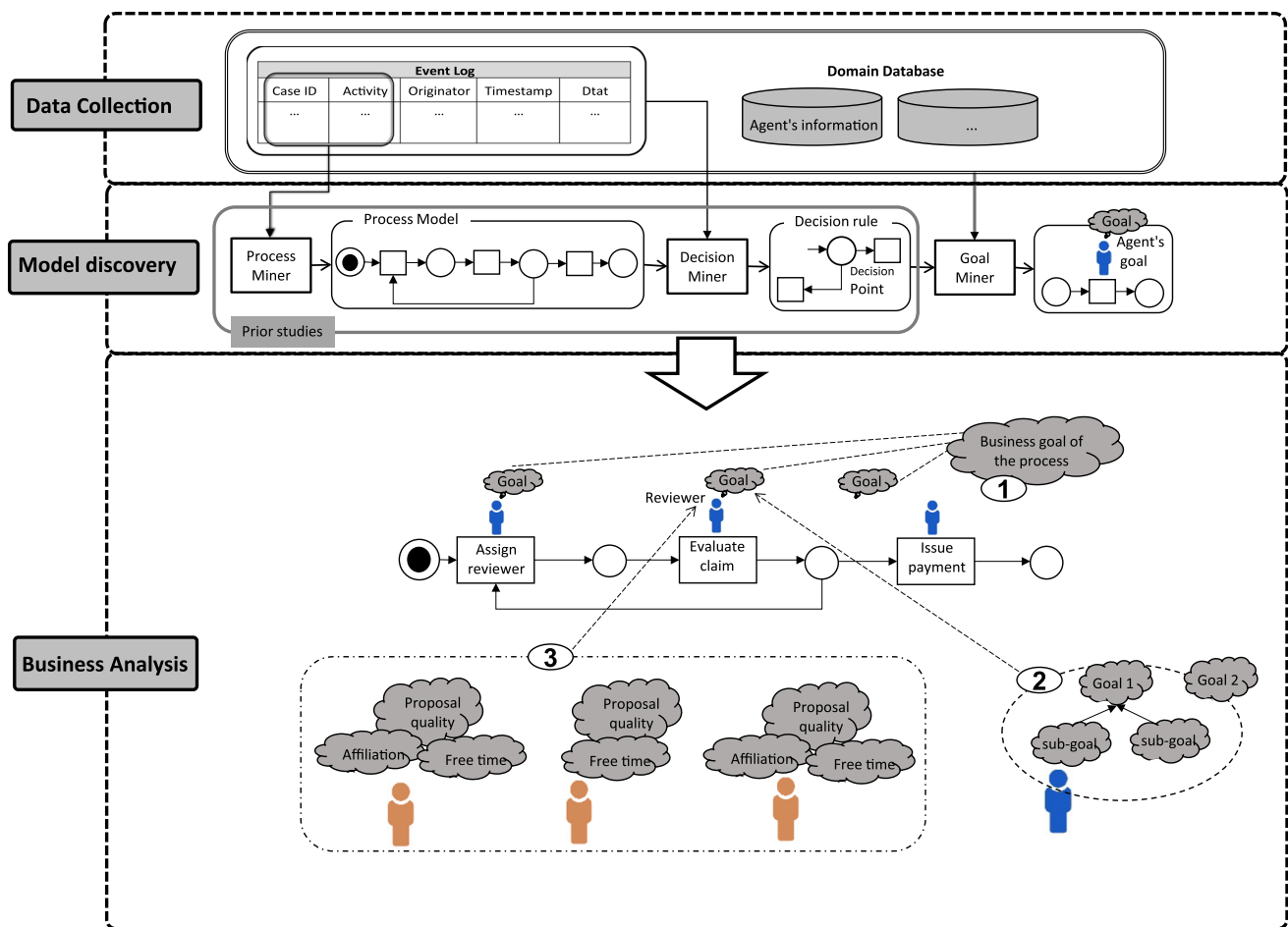


Fig. 3 Yan and Hu's framework proposing an agent goal mining approach Adapted from [115]

recognition of the goal that motivates an employee (agent) to perform a particular sequence of tasks. This is crucial to determine what tasks are expected to be executed next in order to achieve the goal within the dynamics of the organization.

Traditional process discovery techniques [20] do not focus on detecting the goals that motivate actors to do their tasks. Nevertheless, the line of work of Armentano and Amandi [7] corresponds to the area of *goal recognition*. In this direction, they considered the prediction of the subject's goal as an inherently uncertain task. Therefore, they looked for a knowledge representation able to capture and model this kind of uncertainty.

The Markovian approach, proposed by Cook and Wolf [19], is one of the traditional approaches for process discovery. The approach presented by Armentano and Amandi [7] differs from the Markovian approach in that the latter one used Markov chains of first and second orders, while the former one used variable-order Markov models.

In this proposed approach, the model is constructed from the observation of the tasks that an employee

executes and from the goal that motivates the execution of those tasks. The model predicts the most probable agent's "class" (*goal*) after each performed activity. A learning algorithm is used to make a model of the tasks necessary to achieve different goals. This knowledge could be preserved in the organization, contributing to the organizational memory. Finally, this can contribute in detecting the goal of an employee at any time and can help identifying deviations from expected behaviors [7].

Normally, any industry-scale business process takes different variants that ideally contribute to achieving the goals of an organizational goal model. Consequently, the adequate management of process models with many variations mandates considering each process variant as an independent model entity. Ponnalagu et al. [81] worked on the admission of a deviating process instance as a valid variant of the intended process that achieves the same goals as the intended process. They focused on concept and goal conformance drift that rises with evolutionary changes in process executions. This could be admitted as

a crucial aspect especially in knowledge-intensive processes, where significant drifts from valid (goal aligned) workflows could happen due to manual errors and environmental constraints.

There exist studies that worked on the categorization of processes [70], discovering a family of process variants from a collection of event logs [15], and on the management of large collections of process variants of a single process model [42]. In their approach, Ponnalagu et al. [81] validate and categorize the discovered family of variants based on a goal model. Their approach can leverage the works that focus on enriching process designs with goal-driven configurations [65]. Categorizing process execution deviations in a goal-based fashion is necessary to decide whether a deviation represents a valid variant. Ponnalagu et al. [81] proposed an approach to help with the decision of whether process instances in execution logs are valid variants, using a goal-based notion of validity. Their approach also supports the analysis of the impact of contextual factors in the execution of specific goal-aligned process variants. They used the KAOS methodology for goal decomposition. Moreover, they examined their approach with an eclipse-based plug-in and, to validate their approach, they considered a process log of 25,000 events obtained from the history of a help desk division in an IT organization. They demonstrated that with semantic annotations of process models, goal models in terms of the end effects are accurately applied for a given domain. Also, in terms of being the right candidate for dealing with a larger data set composed of processes, their approach is general enough, i.e., it is not restricted to goal models or process models of any scale or domain.

As explained previously, *conformance checking* is acknowledged as one of the main categories of process mining activities, where the real executed process is compared to the process model in order to find and analyze deviations. There are many conformance checking approaches mainly focusing on aspects of business processes, but this is not adequate for analyzing whether the actual business processes are able to satisfy organizational goals.

According to Horita et al. [45], goal-oriented requirements analysis methods received notable research interest in requirements engineering. This approach is useful for reflecting organizational requirements into business process models, but actual business processes usually deviate from defined process models. Hence, it is important to analyze the data recorded as event logs along with using the model's information. Horita et al. [45] proposed a goal-oriented conformance checking approach that not only can detect deviations between logs and models, but can also analyze the effects of a deviation.

As shown in Fig. 4, the method proposed by Horita et al. [45] consists of two main phases. The first phase, called

trace and goal processing, checks whether event logs satisfy the goals of a goal model represented as logical formula and constructs cross-tabulation tables for the second phase. All traces are divided into goal satisfied traces or not satisfied traces. The numbers of traces when a goal is or is not achieved compose these cross-tabulation tables. The second phase, called statistical analysis, considers the relation between the goals and analyzes the positive effects or negative effects of a violated goal on the other goals in the goal model. This phase uses statistical hypothesis testing on cross-tabulation tables constructed in the first phase as an input, in particular Chi-squared test and Fisher's exact test. The authors have applied their approach on an event log of a phone repair process.

In [11, 12], Bernard and Andritsos focused on *Customer Journey Mapping* (CJM) as an emerging research area. In particular, CJM tracks and describes customers' responses and experiences when they are using a service and represents them, as *customer journeys*, on a map [11]. The challenge with CJM is that it is not clear how to present thousands of customer journeys. Inspired by process discovery techniques, borrowed from process mining, Bernard and Andritsos developed the CJM-explorer (CJM-ex) tool. This tool is a web interface that enables interactive navigation through several journeys stored in event logs. To this end, the application uses hierarchical clustering, statistical indexes and *user-defined goals*. CJM-ex aims to show how the event logs can be displayed on CJMs, and to let analysts navigate through these journeys in a goal-oriented fashion.

In the proposed method, a hierarchical clustering algorithm is first used to segment the original customers' traces into groups of similar journeys. After forming the clusters, CJM-ex becomes able to leverage the contextual information (e.g., emotions or characteristics) associated with a typical customer journey. The clustering algorithm generates a tree (or dendrogram) where the journeys seen in event logs are at the leaf level and get merged at higher levels to form "*representative*" journeys.

From a goal-oriented perspective, this method does not deal with customers' goals or intentions. However, it lets the users of the tool (e.g., analysts) define their own exploration goals. For example, an analyst might be interested in journeys that started by a specific activity (e.g., "calling the customer service") experienced by customers with specific characteristics (e.g., people younger than 16). Based on the goal setting, important branches of the clustering tree will be highlighted in "hot" areas of the tree. This feature makes CJM-ex the first goal-oriented tool that enables analysts to set a priori goal to guide journey exploration.

As explained in Sect. 2.2, *goal models* play an important role in requirements engineering by representing statements of stakeholders' intent in a hierarchically form. Here, the goals in higher levels of the hierarchy (parent goals) are

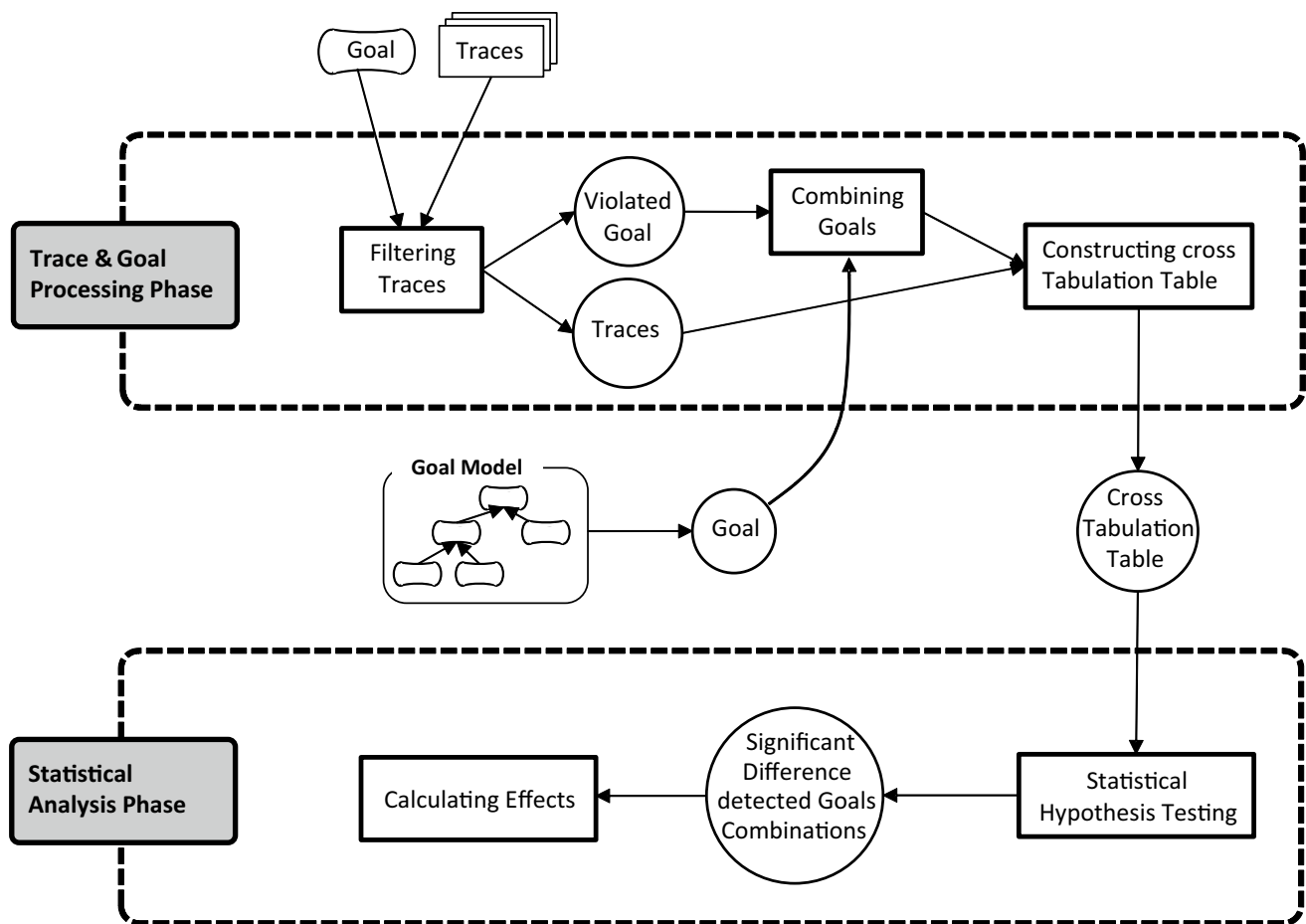


Fig. 4 Overview of the goal-oriented conformance checking method proposed by Horita et al. Adapted from [45]

related to goals in lower levels (sub-goals) via AND or OR refinement links. In that context, Santiputri et al. [92] aimed to address the question: *can enterprise goal models be mined from readily available enterprise data?* Note that the method they proposed does not consider mining *goals* directly from event logs. Rather, it aims to leverage operational data that manifest pre-deployed goal refinements. In summary, this method involves mining event logs to leverage so-called *temporal correlation patterns* between goals and sub-goals, mining goal refinement patterns from multi-layered event logs and composing goal refinement patterns to obtain goal models (or goal trees). Temporal correlation patterns indicate that the achievement of a goal relates to the achievement of its sub-goals. Such a pattern requires that in a sequence of events, the events denoting achievement of a parent goal occur immediately (or soon) after the events denoting the achievement of its sub-goals. For example, the event that denotes occurrence of “making a cup of tea” (as a parent goal) occurs immediately after the events that denote “putting a teabag in a cup” and “pouring of hot water into the cup” (as its sub-goals).

In this approach, it is critical to assume that the input event logs are partitioned into different levels of abstraction. This key assumption suggests that the events implying achievement of parent goals appear in the log of more abstract events (or higher level), while the events implying achievement of sub-goals appear in the log of more refined events (or lower level).

The next step is to compose goal refinement patterns to obtain goal models. Here the main challenge is the difficulty in relating semantically similar, but syntactically highly dissimilar requirements of goals and sub-goals. For example, a sub-goal may be represented as: *log labor hours for billing*. Pretty distinctly, we might mine a goal refinement pattern for a parent goal represented textually as: *track technician time for charging the customer*. Human intuition advises that these two goals are semantically similar, and any existing know-how for the latter is also useful for the former. To cope with this challenge, Santiputri et al. use a state-of-the-art mechanism for measuring syntactic and semantic similarities of words and phrases, called *word2vec* [72]. This mechanism consists of a two-layer neural network

that returns a real-value measure of semantic similarity of two given words (the higher the value, the more similar). Hence, using an appropriate threshold defined by domain experts, one can relate a phrase describing a sub-goal in one goal refinement pattern to a phrase describing a parent goal in another goal refinement pattern. This approach does not guarantee the accuracy of all mined goal models and hence analysts' oversight and editing are still required. This method helps modelers improve their efficiency as they can use the "first draft" models or model fragments (instead of a "blank sheet") to design usable models.

This approach was evaluated using both real-life and synthetic datasets. Overall, the results of the empirical evaluation suggested that the combination of two proposed techniques (sequential pattern mining for leveraging temporal correlations patterns and word2vec for evaluating goal/sub-goal similarity) is a promising basis for goal model mining.

In [22], Dabrowski et al. took into consideration a common problem of software designers during the elicitation of software requirements and propose a solution based on a goal-oriented process mining approach. Software designers have to make assumptions (often invalid) about the users' processes and consequently about the requirements supporting such processes. Elicitation and validation of such assumptions manually through interviews and observations is time-consuming and expensive. Furthermore, designers may fail to find the users' real processes. To address such a problem, Dabrowski et al. [22] proposed a new approach that leverages the synergy between *requirements engineering*, *crowdsourcing* and *process mining*. Such a combination is employed to identify and validate software process requirements exploiting the crowd's perception [40, 93]. This approach takes advantage of the capabilities of process mining to discover the underlying processes of crowds from event logs. A crowd brings an opportunity to involve large and diverse groups of users to interact with a given system and conduct their intended processes leading to achievement of their goals. Then, process mining techniques can discover such underlying processes from the logs stored in databases. The discovered processes become implicit feedback used for revising the existing system's functionalities and for ensuring that a software system operates as expected. In addition, missing functionalities and quality issues may be revealed by the analysis of user–system interactions.

The novelty of this approach pertains to its ability to extend requirements engineering techniques by revealing new requirements using process mining techniques along with crowdsourcing. This is different from other crowdsourcing techniques in the literature that are mainly focused on extracting explicit user feedback [40, 69].

The work of Dabrowski et al. [22] aims to answer the question "how to discover requirements through goal-driven process mining?". They explained their ideas through a

simple example (a personal energy management system) as a case study and illustrated the motivation behind the approach. First, end-users execute their daily processes using the application for a given period of time. Then, the corresponding usage log is input to process mining algorithms to extract the underlying process models. The purpose of this process mining step is to reveal typical patterns of user behavior, process bottlenecks and misalignments or variants between reality and intended nominal processes. This typically benefits a software engineer that has problems detecting discrepancies between real processes and designed ones [90]. Such a benefit is possible by conformance checking and denoting the points in the process not aligned with the high-level user goals [21]. This step results in a list of possible new requirements to address misalignments between designed and actual processes. Further analysis here could lead to the identification and/or confirmation of new stakeholder goals and underlying requirements. The resulting artifact then forms an input for the *planner* who is responsible of identifying tasks and objectives to be delegated to the *crowd*. In the proposed approach, "crowd" denotes a large and diverse group of users delegated to perform some processes using the given application to accomplish predefined objectives [40]. The planner then orchestrates objectives and tasks to be carried out by the crowd.

The crowd interactions with the software are similarly stored as logs and subsequently analyzed. If a misalignment is still present, then this may lead to the implementation of newly found requirements in the application. The synergy between crowd and process mining techniques can indeed become an important source for new requirements.

The work of Dabrowski et al. [22], which exploits users' real behaviors for detecting misalignments, can effectively contribute to requirements engineering and, also, to process improvement. This contribution is promising not only in software systems but also in organizations striving to improve compliance and design better processes.

In this regard, Outmazgin and Soffer [75] proposed a process mining-based analysis to study *intentional incompliances*, where employees intentionally deviate from the required procedures even if they are aware of them. Understanding intentional incompliances, or *work-arounds*, can improve an organization's performance by helping redefine requirements and redesign processes and support systems.

There exist numerous research activities about compliance conducted separately within the communities of requirements engineering [36, 71, 117] and process mining [27, 68, 98]. Existing process mining techniques for compliance checking do not distinguish intentional incompliance and do not address their reasons and sources. In contrast, Outmazgin and Soffer [75] defined a list of generic work-around types found in real practices. They discussed whether

and how these work-arounds can be detected by process mining, and they analyzed their sources.

A work-around denotes the employees' perception of prescribed procedures as an obstacle for some of their desired goals, when intentionally not following these procedures. Work-arounds are usually recognized as a negative phenomenon, assuming that the prescribed process is designed and optimized to reach the desired business goals. However, since work-around is intentional behaviors of employees, they are performed for certain reasons. Poelmans [78, 79] reminds that users work-around a system for the purpose of saving time and/or efforts or avoiding the system's limitations. Outmazgin in [74] shows that work-arounds may be triggered when the predefined business processes are not able to accommodate atypical situations that may arise. Moreover, work-arounds can be motivated when the designed process or its support system cannot satisfy all the stakeholder expectations. Other cases might occur when employees decide to pursue their own personal goals rather than to follow the process designed based on the organization's goals.

The qualitative study of Outmazgin [74] that explores work-arounds in several organizations led to six generic types: A) bypass of process parts, B) selection of an entity instance that fits a preferable path, C) post-factum information changes, D) incompliance to role definition, E) fictitious entity instances and F) separation of the actual process from the reported one. Then, in [75], Outmazgin and Soffer investigate whether and how the work-arounds can systematically be revealed using event logs. To this end, they have used Disco [33] and explored the logs of five processes belonging to three organizations over two years. They further characterized the log patterns that can be used to detect work-around of types A, C, D and F. Also, they explained why such pattern cannot be defined for the other two types (B and E). For example, an experienced employee splits purchase requisitions and places several small requisitions, instead of a high-priced one, to avoid long approval paths [75].

Next, Outmazgin and Soffer [75] used process mining to conduct a quantitative analysis of situational reasons that can be associated with work-arounds. Using the data of real processes, they have statistically tested the hypothesis of a correlation between different types of work-around and *activity durations*, *number of participants* and *work handover*. For instance, they found that work-around of type A (bypassing some activities), as the most frequent type, is positively correlated with activity duration and with the actual number of activity participants.

Aligned with these results, a case study in Belgium [79], also suggested that the sources of work-arounds can be mitigated through the following strategies: deliberate efforts to decrease specific activity times, improving control and designing appropriate work handovers, enlightening access

permissions to prevent unauthorized employees doing activities, and monitoring work-arounds and performing disciplinary actions. These practices are expected to lead to performance and compliance improvement.

One of the principal domains where process mining capabilities are leveraged is healthcare [39]. In particular, clinical pathway (CP) mining from historical data has received increasing attention. Some approaches focus on using *topic modeling* methods to discover clinical patterns instead of process models. The topics are semantic compositions of clinical activities often discovered using latent Dirichlet allocation (LDA), a statistical model developed for text mining and machine learning first presented by Blei et al. [13]. Xu et al. [113] used topic modeling together with process mining to generate a concise and interpretable topic-based process model. The key principle in their work is to discover the latent topics from clinical data using LDA and, then, to extract the order relations between these topics using process mining techniques. In this approach, treatment activities are mapped to words and the latent clinical topics are mapped to latent topics of texts.

Xu et al. [114] further aimed to consider those latent clinical topics as *clinical goals*. They assumed that the clinical activities taking place in a specific time interval (usually a hospitalization day) are prescribed for multiple clinical goals, and each clinical goal corresponds to a set of clinical activities. Finally, a process discovery framework is applied on the goal-based sequences (instead of the detailed clinical activities) to get a comprehensive process model [114]. In order to evaluate the proposed approach, they used two real-world datasets. The results show that the topics revealed by their method provide higher coherence, informativeness and coverage than the raw LDA. These extracted high-quality topics are proper for representation of the clinical goals. Also, the method is effective in generating a comprehensive topic-based clinical pathway model.

Baek et al. [8] focused on the adaptability of the processes discovered by process mining techniques in terms of being dynamically reconfigured for new requests made on business processes. They proposed an analytical method using a heuristic algorithm. This algorithm is based on the goals to create an adaptive process mining model. The model is able to provide a continuous service demand scenario that is created dynamically. Baek et al. proposed some measurements to calculate the correlation among the activities or the importance of each activity within a given process and the similarity of some processes. They took advantage of concepts analogous to basic ordering relations determined by the α -algorithm [66]. They also used the concept of *foot-print matrix*, also known as *order matrix*. Their preliminary study was mainly aiming some further studies that propose a process-based goal scenario. This approach could be a basis

to set up strategies allowing rapid dynamic reconfiguration to meet demanded requirements. This paper did not involve any standard goal-oriented language in conjunction with a process mining technique.

4.2 Intention mining

As described in Sect. 2.1, process mining techniques mainly focus on event logs, discovered models and predefined process models to extract process-related insights [102]. Hence, process mining has been growing into an activity-oriented approach [25].

A research direction called *intention mining* has, however, recently emerged. Intention mining shares objectives similar to process mining's but here the main goal is to discover *intentional process models*, beyond activity process models. Intentional process models consider the intentions underlying activities rather than the activities themselves. Aligned with goal modeling characteristics, the reasoning behind the activities is specifically addressed by intentional process models. This research field is developed by taking into account the notions of *intention* and *strategies* of the process enactment [52, 54, 55].

4.2.1 From intentional process modeling to intention mining

While *intention mining* is quite a young field, *intentional process modeling* (with MAP) as a basis for intention mining goes back to the late 1990s. According to the trend of this research field presented by Epure et al. [28], many studies in intention-oriented process modeling proved that the fundamental nature of processes is mostly intentional and hence the process should be modeled from an intentional point of view. In the regular research trend of intentional process modeling, event logs have, however, been neglected. This neglect is the motivation for research in intention mining. The main challenge of these studies is to identify and formalize intentions from event logs [28, 52–55].

4.2.2 Process mining versus intention mining

Khodabandelou et al. [54] compare two process mining and intention mining approaches by applying process discovery techniques to find a process model from a case study's dataset of traces and discuss the results. To this end, the α -algorithm [66] and the strategies miner algorithm [54] are applied in process mining and intention mining approaches, respectively.

Khodabandelou [52] has elaborated some aspects of intention mining and described its similarities to and differences from other fields of research such as process discovery approaches in process mining, goal-oriented approaches

in goal modeling languages, machine learning approaches in process mining and recommendation techniques. She emphasized that the process discovery approaches discover process model from sequences of actors' activities, whereas intention mining relies on the intentional aspect of the processes to discover the process model. The intentions are defined as the motivation to achieve the goals. The behaviors of users depend on their intentions and all of the intentions are not detectable. It is important that the intentions can be modeled at different levels of abstraction. A model of high-level intentions represents parent intentions and a low-level model shows sub-intentions. The fulfillment of a parent intention depends on the fulfillment of its sub-intentions.

Also, the objectives, representation languages, algorithms and tools of both fields are briefly introduced by Khodabandelou et al. [54]. As described in Sect. 2.1, there are many process mining algorithms and some concrete tools based on them, but the algorithms of intention mining are yet limited to the *hidden Markov Model* (HMM) method (described later). Furthermore, there is no intention mining tool that extracts intentional process models and infers goals from activities.

4.2.3 Language selection for intention mining activities

In order to choose a goal modeling notation to be used in intention mining, Khodabandelou et al. [54] compared three goal modeling approaches (i^* , KAOS, and MAP). The approaches were compared in terms of their variability for goals, rigidity of task-decompositions and their operational semantic of tasks. They rejected i^* because " i^* is not designed to be a variable framework; therefore, it does not afford a high level of flexibility" [54]. However, this argument deserves to be challenged since i^* takes *intentional* and *strategic* views and supports representation of variability through means–ends relationships, enabling to perform both "upward" and "downward" strategy evaluation and to analyze the types of variability [116]. KAOS was also rejected because it has a rigid task-decomposition mechanism, and hence modeling complex intentional processes would be difficult. In addition to the aforementioned modeling language, Khodabandelou [52] has considered GRL and Tropos as modeling languages and did not choose them as they are based on i^* and thus were considered to have limitations similar to i^* 's.

MAP (introduced in Sect. 2.2) was chosen because this notation represents task and strategies in a labeled directed graph (aligned with the intention mining techniques) and allows formalizing flexible processes [24, 52].

4.2.4 Hidden Markov models as a basis for intention mining

One cannot easily understand HMM-based intention mining methods unless he/she has a good knowledge of HMM, e.g., as tutored by Rabiner [85]. Addressing this important required knowledge, an introduction to HMM is presented here as a prelude to its usage in the upcoming description of existing intention mining approaches.

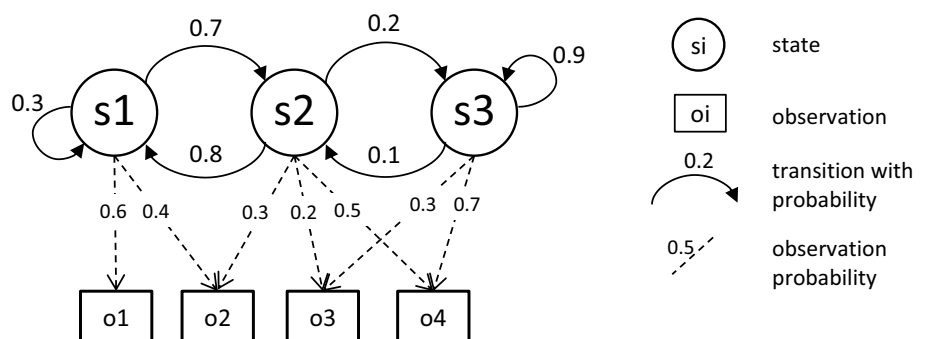
A *hidden Markov Model* (HMM) is a statistical model that consists of two main components: *states* and *observations*. The state of the model in any given time generates an observation in that time. For example, in a meteorology context, states are the weather types (sunny, raining, etc.) and the possible observations generated with the states can be walking, shopping, staying at home, etc.

Figure 5 shows a simple HMM with three possible states and four possible observations. At a given time t , the state, s_t , is a member of the set of states called $\mathbf{S}$. In Fig. 5, $\mathbf{S} = \{s1, s2, s3\}$. State, s_t , depends on the previous state, at time $t - 1$; the system goes from state, s_i to state, s_j with a probability called T_{ij} . Note that the sum of the probabilities of a given state's outgoing transitions is 1. For example, in Fig. 5, $T_{12}=0.7$, $T_{11}=0.3$ and $T_{13}=0$ (0 probabilities are not shown in such models). Therefore, in state $s1$, the probability of moving to state $s2$ is 0.7 while it is 0.3 to state $s1$.

Visiting a state at a given time can generate an *observation* from set $\mathbf{O}$, with a given probability. In our example, the set of possible observations is $\mathbf{O} = \{o1, o2, o3, o4\}$. Let E_{ik} be the probability of generating observation k in given state i . In Fig. 5, state $s1$ can generate observations $o1$ with a probability of 0.6 (E_{11}) and $o2$ with a probability of 0.4 (E_{12}). Let the numbers of possible states and observations be N and M , respectively. Accordingly, there are two matrices that represent all T_{ij} and all E_{ik} , respectively, called transition ($\mathbf{T}: N \times N$) and emission ($\mathbf{E}: N \times M$) matrices. For instance, the transition and emission matrices for the example in Fig. 5 are:

$$\mathbf{T} = \begin{bmatrix} 0.3 & 0.7 & 0 \\ 0.8 & 0 & 0.2 \\ 0 & 0.1 & 0.9 \end{bmatrix} \quad \mathbf{E} = \begin{bmatrix} 0.6 & 0.4 & 0 & 0 \\ 0 & 0.3 & 0.2 & 0.5 \\ 0 & 0 & 0.3 & 0.7 \end{bmatrix}$$

Fig. 5 A hidden Markov model with three states and four observations



In addition, the vector π contains the probabilities of each state to be the initial state. In our example, $\pi = (0.9, 0.05, 0.05)$, which means that the system starts from $s1$ with a probability of 0.9 (the π values are not shown in the figure). The transition and emission matrices, together with the initial vector, capture the entire model and can be formalized as a triplet, called $\lambda = (\mathbf{T}; \mathbf{E}; \pi)$. The main characteristic of HMM is that the states are not observable and remain hidden whereas the observations are the only things visible to users. Consequently, there are three key and pragmatic problems that should be solved, where A is a sequence of observations, e.g., $A = (o1, o1, o2, o4, o3, o3)$:

- Problem 1: given the sequence A and the model $\lambda = (\mathbf{T}; \mathbf{E}; \pi)$, what is the probability of A , i.e., $\Pr(A | \lambda)$?
- Problem 2: given the sequence A , what is the optimal sequence of states that could generate A ?
- Problem 3: given the sequence A , how can we estimate the model parameters, $\lambda = (\mathbf{T}; \mathbf{E}; \pi)$, that maximize $\Pr(A | \lambda)$? Here, the transition matrix, the emission matrix and the initial vector of probabilities need to be found.

There exist mathematical algorithms to solve the aforementioned problems. The *Forward Algorithm* is the main one for solving the first problem. The second problem can be solved with the *Viterbi* algorithm, which finds the state sequence with the maximum likelihood, given the observations sequence. The third problem is, however, the most difficult one, as there is no known analytical way to find the most likelihood model. However, there is an iterative method, known as the Baum–Welch Algorithm (BWA) [10], that can find a local maximum likelihood estimate of the parameters given a set of sequences of observations [84].

4.2.5 Map Miner Method (MMM), unsupervised and supervised

The Map Miner Method (MMM) aims to infer a MAP intentional process model from event logs. If a prescribed MAP already exists, an analyst can compare the discovered MAP

with the prescribed one and study the compliance level and deviations. In this sense, the MMM approach shares similarities with process discovery techniques in process mining. The output of process discovery is a discovered process model in the level of activity, while MMM ends up with a model in the intention level. Intentional MAP can be inferred through either an unsupervised or a supervised scheme. Here, we describe these two methods, which are illustrated in Fig. 6.

In the work of Khodabandelou et al. [57–59] the focus is on *unsupervised* methods. They took advantage of HMM to propose a MMM framework to find an intentional MAP from event logs. In this framework, the hidden *states* are *strategies* and the *observations* are users' *activities*. Since activities are generated by users' strategies, the activities that are observable in event logs would be generated by the strategy that the agent has adopted to achieve an intention. Therefore, the

sequence of agents' strategies will generate the sequences of activities stored in the event log. For example, as Fig. 7 shows, the trace of activities $\langle a, b, b, c, d \rangle$ observed in event logs could be generated by different traces of strategies such as $\langle s1, s1, s2, s2, s3 \rangle$ or $\langle s1, s2, s2, s2, s3 \rangle$.

Generally, each sequence of activities can be generated by different sequence of strategies with respect to the transition and emission matrices. As Fig. 6 shows, in the unsupervised MAP Miner Method, the traces of activities are used as inputs. Although the strategies are some groups of activities, here, there is no previous knowledge about the strategies and the segmentation of activities' sequences into strategies. Therefore, in addition to the transition matrix (going from one strategy to another one), the emission matrix (generating an activity by a strategy) is also unknown. As the activities in the traces are not labeled by corresponding strategies, this method is called *unsupervised*. Here, the

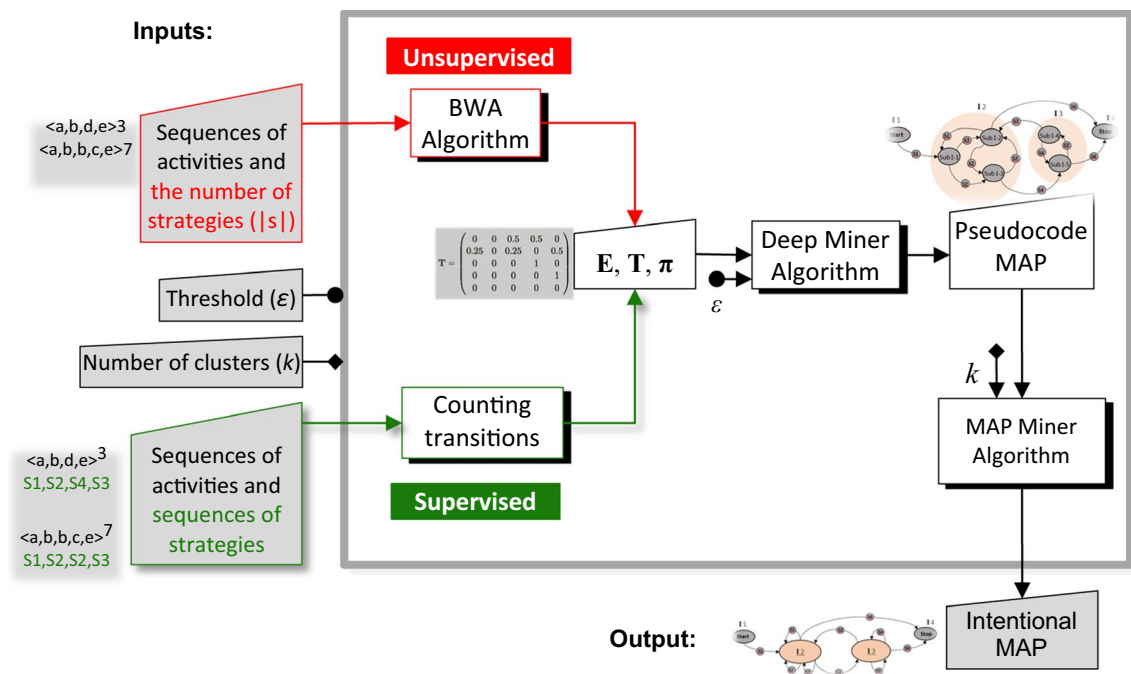
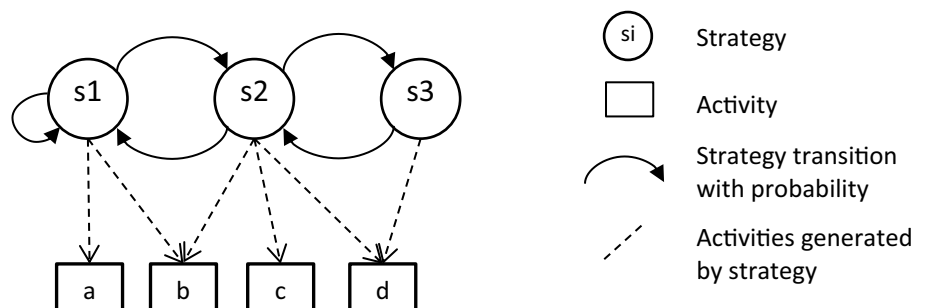


Fig. 6 The supervised and unsupervised Intentional Map Miner Methods, with their inputs, algorithms and outputs. The red items belong to the unsupervised method only, and the green items to the supervised method. The black items are part of both methods

Fig. 7 A sample hidden Markov model that consists of strategies as hidden states and of activities as observations



observations are known and the $\mathbf{T}$ and $\mathbf{E}$ matrices should be estimated. The HMM parameters are estimated using the Baum–Welch Algorithm (BWA) [10], the most frequently used algorithm for learning the parameters of a HMM (discussed in Sect. 4.2.4).

In addition to the observations, the BWA algorithm also needs the number of states as input. Therefore, the number of strategies should be defined by an expert in the unsupervised method. The main outputs of the BWA algorithm are the transition and emission matrices (see Fig. 6). The transition matrix shows the probability of moving from one strategy at time t to another strategy at time $t + 1$. Once $\mathbf{T}$ is estimated, the problem becomes how to extract a MAP process model (described in Sect. 2.2) that reflects the transition matrix well. The MAP extracted from a transition matrix should be verified in terms of two constraints: (1) *fitness*: any transition between strategies possible in the transition matrix should also be possible within the MAP; (2) *precision*: any transition that is possible between strategies in the MAP should also be possible in the transition matrix. Deneckère et al. [24] and Khodabandelou et al. [58, 59] introduced *recall* and *precision* as metrics of fitness and precision, respectively.

The main assumption here is that such unsupervised approaches only consider the transitions whose probability is above a given threshold ε . The value of ε is chosen

heuristically to obtain a suitable trade-off between the granularity of the MAP and its understandability. Hence, ε is considered as an input of the algorithm in Fig. 6. The F-measure metric, a combination of both recall and precision, is used to qualify the MAP. Deneckère et al. [24] proposed an algorithm, called *Deep Miner*, which finds a MAP that maximizes F-measures with the lowest number of sections (see Fig. 6). The output of the deep miner algorithm is a very detailed map with several nodes, called *fine-grained* MAP. Figure 8 shows a *fine-grained* MAP extracted by the deep miner algorithm in an example with six strategies. The nodes generated by the deep miner algorithm are called “*sub-intentions*” as they are connected to each other with some strategies.

Deneckère et al. [24] and Khodabandelou et al. [53, 57–59] also proposed a MAP miner algorithm to extract a high-level MAP from the fine-grained MAP produced by deep miner. To this end, the MAP miner algorithm uses a clustering algorithm, *K-means* [44], to group the sub-intentions into some higher-level nodes (as clusters). In the clustering step, the attributes of each given sub-intention are defined as binary values that represent the connectivity of that sub-intention to the other nodes. Note that in the K-mean clustering algorithm, the number of clusters (k) has to be chosen in advance. Accordingly, the parameter k ,

Fig. 8 A fine-grained map extracted by deep miner algorithm includes six strategies [53]

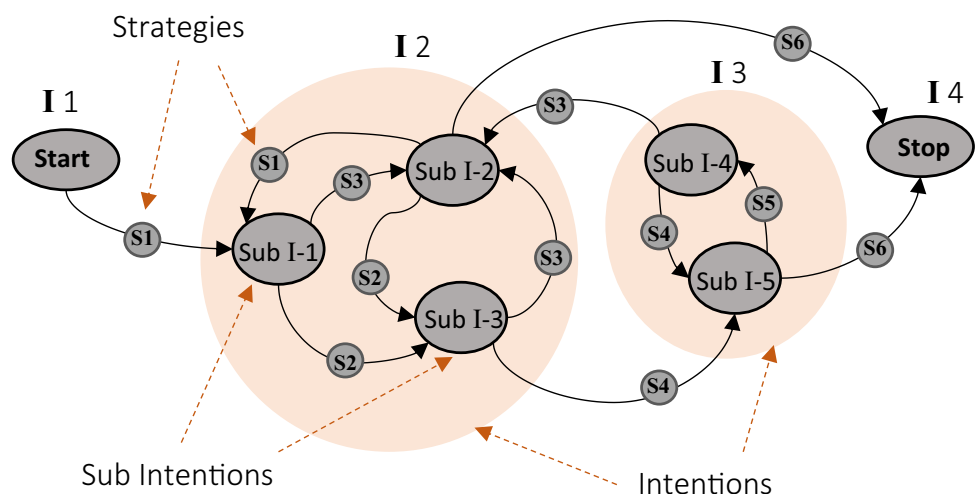
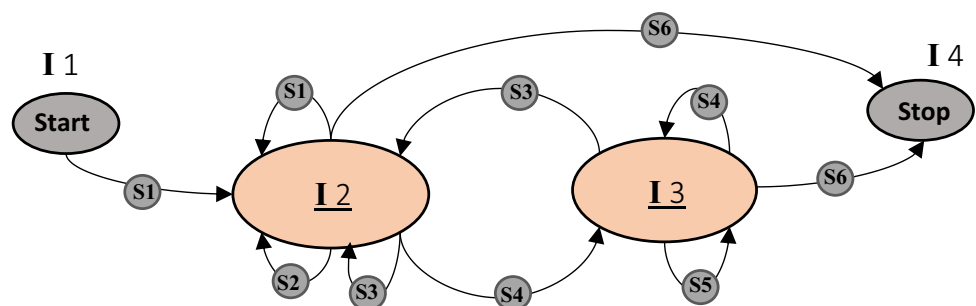


Fig. 9 A coarse-grained MAP generated from the fine-grained MAP in Fig. 8, using the MAP miner algorithm [53]



which is the number of intentions in the final process model (coarse-grained MAP), is another input of MAP miner. This allows determining the level of abstraction for the final intentional MAP. The MAP miner algorithm groups the five sub-intentions of Fig. 8 into two high-level intentions (I2 and I3), as shown in Fig. 9.

In the *supervised* approach, proposed by Deneckère et al. [24], the steps are similar to the ones for the *unsupervised* approach, but the inputs are slightly different. In the supervised approach, the sequences of strategies (not just their number) are also known. For example, in the model shown in Fig. 7, the sequence of activities (e.g., $\langle a, b, b, c, d \rangle$) and the corresponding sequence of strategies (e.g., $\langle s1, s1, s2, s2, s3 \rangle$) are known.

In hidden Markov models, when the sequences of observations and their corresponding states are known, estimation of emission and transition matrices is simple. This problem, called HMM learning, is not one of the tree aforementioned problems of HMMs. Here, nothing is really hidden. When the underlying states are known, a maximum likelihood estimation approach estimates $\mathbf{T}$ and $\mathbf{E}$ to have the most likelihood of generating the sequence of observations and the sequence of states simultaneously. In this approach (called *counting transitions method*), T_{ij} is the number of transitions from state i to state j divided by the number of times that state i occurs. Also, E_{ik} is the number of times that observation k is generated by state i divided by the number of times that state i occurs.

As shown in Fig. 6, in a supervised approach, the BWA algorithm is not needed and the counting transitions method can simply be applied to estimate HMM parameters. After this point, all the steps are similar to the unsupervised approach and the MAP intentional model will be finally extracted. It is worth noting that the number of clusters (k) in the MAP miner algorithm does not have to be chosen in a supervised approach as it equals the number of strategies that are known [55].

4.2.6 Comparing the performance of supervised and unsupervised approaches

Khodabandelou et al. [56] compared supervised and unsupervised learning approaches of HMMs in both a theoretical context (in terms of convergence speed and likelihood) and a

practical context (in terms of the intentional process models obtained). The results showed that supervised learning ends up having a poor performance because it enforces binding conditions in terms of data labeling, introduces inherent human biases and delivers unreliable results in the absence of ground truth. On its side, unsupervised learning gets efficient MAPs with a better performance and lower human effort. Rabiner and Juang [84] also demonstrated that an unsupervised learning approach to estimate HMM parameters offers a higher performance than supervised learning.

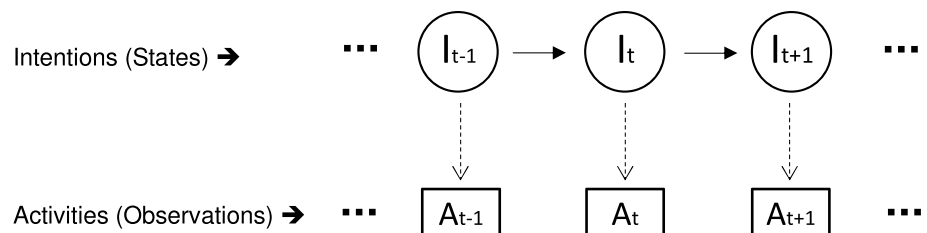
4.2.7 Supervised intentional process model discovery using HMMs

Khodabandelou et al. [55] proposed an intention mining method called supervised intentional process model discovery using HMMs. This approach is similar to the supervised MAP miner method, described in Sect. 4.2.5, but here the states of the hidden Markov model are *intentions* instead of *strategies* (see Fig. 10). This approach looks for some predefined intentions behind the user activities traces and compares them to a prescribed intentional model that is represented with MAP.

In this method, there is a prior model called *intentional process model*. This model is a MAP that consists of some strategies and some intentions. A human expert describes this model. Users can select the alternative strategies to fulfill their intentions, following this intentional model. Then, a modeler makes some groups of activities based on his/her inference to define some strategies that are shown in the prescribed intentional model. Similar to the supervised MAP mining method, this approach adopts hidden Markov models to discover a MAP from observations. Consequently, regarding the characteristics of HMMs, a sequence of activities (traces) of a given length could be generated by many sequences of intentions with the same length. Here, the purpose is to find one sequence of intentions among all possible ones that is the most likely sequence of intentions, which generates the given sequence of activities. This problem is exactly the second key problem of hidden Markov models described in Sect. 4.2.4.

VA [34] is a well-known algorithm for finding the most likely sequence of hidden states that generates a sequence of observed events in hidden Markov models. This algorithm

Fig. 10 The HMM used in the method of Khodabandelou et al. [55]



needs transition and emission matrices. When providing these inputs, the modeler estimates the transition matrix by the simple counting approach described before (in the supervised MAP mining approach). Once the parameters of the HMM are estimated, they can be used to find the hidden intentions behind any sequence of actions traces (the sequence with the highest probability of emergence), using VA. Then, this discovered sequence of intentions could be compared to the prescribed sequences of intentions by a modeler. At this point, Deneckère et al. [24] realized that this method is not accurate enough to represent a MAP model. They argued that activities are indeed generated by strategies, not directly by intentions. Accordingly, Deneckère et al. [24] proposed their supervised Map Miner Method, described in Sect. 4.2.5, which models hidden states as strategies and observations as activities. Note that the supervised Map Miner Method infers high-level intentions by clustering the sub-intentions extracted from the Deep Miner algorithm (shown in Fig. 9).

4.2.8 Unsupervised intentional process model discovery using IntentMiner

Based on the intention mining approach proposed by Khodabandelou et al. [55], Epure et al. [29] proposed an approach, called *FlexPAISSeer* that focuses on the difficulties of unexperienced actors when *enacting* flexible processes. An unexperienced process actor, who is not much aware of the process, may have some difficulties to make a good decision about the action to execute next under specific constraints. Addressing this problem, *FlexPAISSeer* offers two components—*IntentMiner* and *Intent-Recommender*. The first one discovers the intentional model in an unsupervised manner, then the second one makes recommendations based on the discovered intentional model and probabilistic calculus. In this unsupervised approach, the clusters of events that are associated with an intention are identified as a basis for the intentional model.

Epure et al. [29] aim to discover the intentional model through the traces of the process participants. To this end, they mine both the intentions and the flows between them. They first focus on the identification of the relevant data in a specific structure. Then, they follow the *IntentMiner* algorithm and its six steps: extract the process participant current log, sort the log by time, apply syntactic analysis, apply trend analysis, apply semantic analysis and aggregate intentions. While Khodabandelou et al. [57, 58] used BWA to find clusters of the strategies, Epure et al. [29] proposed an algorithm to cluster the events in some classes and analyze them as intentions. The main idea of their clustering is based on the similarity and confidence between the consequent events that are sorted by time. The attributes of events

are the main basis for finding the correlation between them. Once the intentional clusters are determined, the next step is the intention mining and naming by applying semantic analysis. Finally, the algorithm aggregates the mined process instances in the intentional process model. The intentional process model is further used by *Intent-Recommender* for providing up-to-date recommendations to the process participants.

Epure et al. [29] evaluated *IntentMiner* in an experiment with 10 participants, interacting with a childcare system developed by a software company and used by several child day care centers in The Netherlands. They compared the intentions classified by *IntentMiner* with the real intention of participants. Following a confusion matrix classification, the average precision was 69% and recall was 97% (*IntentMiner* mined the process participants' intentions in 97% of cases). Khodabandelou et al. [55] reported an average precision of 0.97 and average recall of 0.93 for their supervised intention mining technique based on Hidden Markov Models. The precision was much better given the fact the classifier had been trained earlier.

4.3 Key performance indicators (KPIs)

Based on the definition of Flapper et al. [32], performance is the degree to which an organization accomplishes its objectives. All performance indicators (PIs) should be linked to the organization's goals [32]. The PIs that reflect the critical or “key” success factors of the organization and are used to define and measure progress toward the organizational goals are called KPIs [32, 49]. The fact that KPIs are known means of monitoring goals is an important motivation to take KPIs into consideration in this literature review. Some goal modeling languages such as GRL [50] and the Business Intelligence Model [47] even support indicators as a core concept. The studies that focus on the use of event logs in monitoring the KPIs are worthy of attention, especially if these KPIs can be connected to goals.

As discussed earlier, the analysis of misalignments between an event log and a prescribed process model is a good source of improvement in requirements. Process mining techniques (i.e., conformance checking and process enhancement) make it possible to diagnose such deviations. The severity of each deviation can be *quantified* to form a basis for tackling the problem of repairing a process model such that the enhanced model can replay the log (as a history of actual behaviors) [30, 103]. Since a process model may deviate from a log at an arbitrary number of points, making such alignment may be very far from trivial. van der Aalst et al. [103] have demonstrated characteristics, limitations and methods for conducting conformance checking in the context of process mining. In this same context, Fahland and

van der Aalst [30] have investigated the problem of repairing a process model to replay the log and conform to it.

Relying on [30, 103], Dees et al. [23] proposed a methodology that repairs a process model with respect to the behaviors that do not violate any rule and have a significant improvement on the performance level. Involving performance levels that correspond to specific business goals or requirements, represented by KPIs, positions their work as a goal-oriented process mining activity. The basic inputs of the proposed methodology are an event log together with a process model (either discovered or designed manually) represented by a Petri net. The output is a repaired and improved process model ready to impose a better way of performing the business process.

In a nutshell, Dees et al.'s methodology is implemented in three main steps. First, a deviation analysis is performed to determine which deviations have a positive impact on the process performance. To this end, deviations are correlated to a selected KPI stored as a field in event log. Then a set of rules is discovered through a decision tree classification method, where the KPI is a label field. Then for each rule, the corresponding class contains all traces that comply with the rule. In the second step, traces of different classes are repaired to only keep those deviations having a positive impact on the KPI. Then all the logs kept in all the classes are merged to obtain a single repaired event log. Finally, the repaired log is used as a basis to repair the process model. To this end, the process model is revised in such a way that the resulting model is able to replay all the behavior of the repaired event log. In particular, this technique will modify the model only to make the desired deviating behaviors possible. Here, the desired deviating behaviors can be considered as those facts that can reflect some neglected requirements.

Dees et al. [23] assessed the feasibility of their proposed methodology through two case studies. The result showed that the improved process model still adheres to the desired prescriptive model and guarantees enhanced KPI levels.

While Dees et al. [23] took advantage of performance indicators to improve business processes, Cho et al. [17] aimed to assess whether process redesign has been applied and to what extent. They proposed an assessment framework, where the focus is on the process redesign lifecycle phase, which is tightly coupled with process mining as an operational framework to calculate identified indicators.

The framework consists of some indicators to assess whether best practices have been applied. Best practice implementation indicators (BPIs) and process performance indicators (PPIs) are two sets of indicators that assess process improvement resulting from the application of best practices. It is possible to calculate both sets using standard process mining functionalities. Cho et al. also define what data should be recorded during process execution to enable

such a calculation. The proposed framework was evaluated over case studies in a hospital and a tour agency and compared with other approaches in the literature.

Cho et al. [17] defined 17 BPIs for 29 best practices that were previously proposed by Reijers and Mansar [86], such as *customer reduction* and *customer integration*. They also suggested process mining techniques appropriate for computing each indicator. In the case where the BPI information does not exist in the event log, they suggested which supplementary information is needed to assess the implementation of redesigns. In addition, 13 PPIs are suggested to assess the effect of best practices on process performance. These PPIs are based on four process performance measures explained by Reijers and Mansar [86]: time, cost, quality and flexibility. Cho et al. gave a detailed explanation on each PPIs including how to measure them using event logs. Most business process redesign efforts are invested in increasing the efficiency of business processes through improving time-related indicators, e.g., decreasing processing time and waiting time. Calculating time-related PPIs (5 out of their 13 PPIs) using event logs is much easier than calculating quality-related indicators (4 out of their 13 PPIs) because they are not clearly stated in the logs. The latter are rather evaluated by checking the satisfaction of customers, for instance via surveys [43].

Krathu et al. [64] used a bottom-up approach for the identification of KPIs from event logs, business information and process models. This approach, together with a top-down approach for measuring business performance on the strategic level (balanced scorecard, BSC), makes a framework that supports inter-organizational performance evaluation. This framework, proposed mainly for evaluation of inter-organizational relationships, is developed as a plug-in of ProM 6, namely the BSC EDImine plug-in [26].

Knowledge about the BSC method is essential to understand the structure of this framework. In terms of the bottom-up approach that uses event logs for identification of KPIs, they follow two major steps: i) *frequency analysis* and ii) *consideration of KPIs based on the semantics of data elements and message types*. This method has been described in detail by Krathu et al. [63]. It is worth noting that these two papers [63, 64] represented the EDImine plug-in, specifically designed for supporting the business activities and transactions within a network of organizations called electronic data interchange (EDI).

The framework proposed by Krathu et al. [63, 64] was validated with a case study using data from a manufacturing company. This case study showed that the company can evaluate the inter-organizational performance quantitatively against their business objectives. The KPIs derived from this bottom-up process mining techniques together with other statistical techniques enabled the company to evaluate their business performance more accurately.

5 Reflective discussion

Synthesizing what we found from the review of the selected papers enables us to reflect on the state of the art and on potential research gaps.

Results of the queries on both process mining and goal modeling (Sect. 3) yield that potentially synergetic effects achievable by combining goal-oriented modeling and process mining have so far been neglected. Filling this gap can help practitioners of both domains deal with some of their own challenges that are difficult to mitigate separately [37, 38].

The review exposed three main categories of studies: goal modeling and requirements elicitation, intention mining and performance indicators. The *goal modeling* category (Sect. 4.1) focuses on the use of process mining approaches in association with process goals. These studies have been done separately and do not represent a coherent line of research in this context. Nevertheless, they expose a broad area of research at the intersection of the two topics from different viewpoints.

- From an *agent viewpoint*, the goals behind activities of agents who contribute in a process (e.g., employees) are considered using three different approaches. Yan et al. [115] proposed an approach to learn goals of agents by classifying their activities in different situations, adopting a decision tree algorithm. Armentano and Amandi [7] considered recognizing goals that motivate agents to do their tasks using variable-order Markov models. Finally, Outmazgin and Soffer [75] analyzed different kinds of intentional incompliances where employees intentionally deviate from prescribed procedures (work-arounds) to find their sources using process discovery techniques.
- A *process viewpoint* (or case/customer viewpoint) that considers all activities constituting a trace is considered in the first category of papers as well. Ponnalagu et al. [81] proposed an approach to analyze and validate a family of variants of a single process based on a goal model. Horita et al. [45] proposed a method to detect discrepancies between logs and prescribed models and to analyze their effects using a goal-oriented conformance checking approach. Bernard and Andritsos [11] used process discovery in conjunction with customers' journeys and developed a tool to facilitate navigation through many different such journeys in a goal-oriented fashion [11]. Xu et al. [114] used LDA together with process discovery to generate a topic-based clinical pathway, beyond activity-based ones, appropriate as a representation of clinical goals. The adaptability of discovered processes to answer new requests made on business processes was considered by Baek et al. [8]. The above took advantage of some

concepts that primarily belong to the α -algorithm [66], such as *footprint matrices*.

- From an *organization viewpoint*, the overall goals that should be achieved by execution of the business processes are considered. Santiputri et al. [92] propose an approach to discover goal refinement patterns of the goal models considering the sequence of events in multi-layered event logs.

Taking advantage of process mining capabilities is promising to enrich requirement elicitation and management. Dabrowski et al. [22] considered eliciting new requirements using process mining techniques along with crowdsourcing from logs of interaction of end-users with a piece of software or application. Using event logs to find requirements, however, deserves to be more developed not only for software specifications but also for activities performed within organizational and social contexts [38].

In contrast with the first category, the second category of papers (Sect. 4.2) shows a meaningful trace of linked studies on *intention mining*. Khodabandelou et al. [54] introduced intention mining in comparison with process mining. The Map Miner Method can infer intentional MAP models through either a supervised [24] or an unsupervised [57–59] approach. These two approaches were compared in theoretical and practical contexts by Khodabandelou et al. [56] and Rabiner & Juang [84]. Based on the supervised intentional process model discovery [55], Epure et al. [29] proposed an approach that aims to address the difficulties of unexperienced actors to make a good decision about the next action they should do under specific constraints. All the papers from this second category have been published after 2013 and hence represent an emerging field that focuses on the intentions underlying activities rather than on the activities themselves.

The third category (Sect. 4.3) consists of papers focusing on *performance indicators*. There are four papers in this category that linked the event logs to indicators. Two papers proposed a framework specifically for supporting transactions within a network of EDI organizations [63, 64]. The other two papers use indicators to modify models according to desired deviating behaviors and to assess impacts of process improvements. In the literature on goal modeling, the main use of indicators is to measure the degree of fulfillment of stakeholders' goals. Indicators are essentially needed to *evaluate* the goals that constitute a goal model [47]. However, this aspect of KPIs is not considered by the papers selected in this third category. Even recent work from Gurgen Erdogan and Tarhan [41] acknowledges the need to consider indicators linked to goals for effective process mining in healthcare, but only do so informally in their approach and without goal models. Accordingly, the research about goal-oriented process mining associated with performance

indicators, as means to quantitatively measure goal satisfaction, requires further attention.

This systematic literature review is of value to the many researchers working either on goal/requirements modeling or on process mining, as it exposes knowledge about concepts and state-of-the-art techniques that cut across these two siloed research domains. It is our hope that the knowledge about both domains presented here will help researchers consider the potential synergy between goal modeling and process mining. Moreover, this review is valuable to requirements engineers interested in obtaining evidence-based definitions of existing processes, their goals and indicators. This enables them to measure goal satisfaction to assess goal-process alignment and help define relevant requirements for their modification or improvement. Practitioners can further benefit from this paper as it raises awareness about the limitations of current activity-based process modeling and mining approaches and about emerging techniques that attempt to discover goals and intentions behind processes.

Note that three aforementioned categories are not entirely mature yet. Accordingly, researchers who are interested in mining goals from event logs (goal mining) or discovering goal-aligned processes (goal-oriented process mining) are invited to take into consideration process mining techniques combined with the abilities of the goal modeling approaches.

Requirements engineers who practice scenario-based elicitation methods can exploit process mining capabilities to generalize instance-level data logs into models. This allows them to take advantage of such data, which are orders of magnitude larger than the data engineers often use.

6 Threats to validity

The objective of a systematic review is to reveal as many primary related studies as possible to summarize all information available about some phenomenon in a systematic and unbiased manner. There exist several threats to the validity of this particular research and of the papers that were reviewed in the previous section. The validity of a research is concerned with the question about the accuracy of conclusions, i.e., the alignment between research conclusions and reality [77].

This section takes into consideration the threats and the biases that may affect the validity of this work and of the 24 papers selected for review. The main threats are discussed according to two categories: *construct validity* and *internal validity* [31].

6.1 Construct validity

Construct validity refers to the quality of the methodology in terms of being helpful to answer the target research

questions. In this paper, like in other literature reviews, searching for and selecting relevant papers played a crucial role. Note that the research questions were answered based on the papers selected from the set found in the searching stage. Hence, any mistake, weakness and imprecision in the first stage could threaten the accuracy of the answers.

Even though the most frequent and important keywords were used in the search queries, it is possible that some relevant papers that have not used those keywords have not been found.

Another threat is related to the search engine limitations. Ghasemi and Amyot [39] suggested that 22% of the papers related to *process mining* are returned only by Google Scholar. This shows that Google Scholar has a notable contribution of relevant references in the context of process mining. Also, as mentioned in Sect. 3.3, one set of the queries' keywords are related to process mining. On the other hand, as Google Scholar only searches within title or full text, the authors chose to search in titles only (to avoid noise). Consequently, the papers whose titles do not have the queries' keywords were neglected by Google Scholar.

Moreover, all queries' keywords were defined in English. Therefore, another threat that may affect the validity of this review is the possibility of overlooking some relevant non-English papers.

Finally, taking advantage of four search engines, including two generic ones (Google Scholar and Scopus) allows to detect a large proportion of the papers stored in scientific databases. Nevertheless, there might be some relevant papers stored in other databases that are not reachable with these search engines.

6.2 Internal validity

Some other biases and confounding factors can threaten the validity of this study and of the other studies that were reviewed. For example, almost all of the papers related to intention mining came from the same group of researchers. This fact could make an environment open for some biases.

In this paper, one additional threat is that the selection and analysis of papers and the application of the criteria were mostly done by one person (the first author), with a confirmation from the second author. Such issue could be mitigated in the future by having more people do such selection/analysis independently, and then use a consensus-building mechanism.

7 Conclusion

This paper first provided an overview of process mining and of goal modeling. Following the systematic literature review approach, the authors ran queries using four relevant search

engines and analyzed the returned papers in order to answer the research questions.

It was first ensured that there was no literature review specifically related to goal-oriented process mining. Although there are reviews slightly related to this topic, including the one of Papadimitriou et al. [76], none of them has focused on mining goals or requirements from event logs and/or exploiting goal models through process mining activities as it would be expected in a goal-oriented process mining approach. Furthermore, several papers related to goal-oriented process mining were found by our queries and reviewed, but none of them was a literature review.

Regarding question RQ1, general queries with keywords related to both research fields, process mining and goal modeling were defined and run. Through their results, we found that although process mining and goal modeling are growing research topics with considerable amounts of publications, there are only a few rare studies conducted at their intersection. Therefore, this suggests that *goal mining* and requirements engineering techniques that exploit data logs, and goal-oriented processes discovery that exploit goals, can still be considered gaps to be filled between the process mining and requirements engineering communities.

Question RQ2 was about the definition of the main categories of existing approaches, and their main contributions to goal-oriented process mining. In order to answer this question, the studies conducted on the joint goal modeling and process mining techniques were obtained and summarized along three main categories: *goal modeling and requirements elicitation*, *intention mining* and *performance indicators*. Then, we synthesized them to reflect on what has been done in each category. The synthesis does not suggest a coherent line of research about process mining in association with goals. However, a meaningful trace of research work was suggested for *intention mining* as the second category. Moreover, we observed that research about performance indicators (as measures of goal satisfaction) associated with process mining is also sparse.

Goal-oriented process mining promises to make a beneficial connection from event logs, as a major part of the process mining concept, to goals and requirements, as essential concepts of most requirements engineering approaches. This connection opens up new horizons in the context of business process modeling and data analysis to mine, monitor and maintain process goals, process performance and satisfaction levels of stakeholders. To this end, URN, as a standard notation that supports activity-based process models, goal models and KPIs, is an interesting candidate modeling notation that we would like to explore further in the future. This research direction, which would fill the goal-oriented process mining gap highlighted in this paper, could lead to developments that would help practitioners understand and

improve processes [37, 41], manage requirements [38] and support organizations in better measuring and achieving their goals. Another research direction is concerned with the mining of declarative process models (rather than procedural ones), which can also be aligned with intentions and goals [94].

Acknowledgements This work is funded by the Discovery grant program of the National Science and Engineering Council of Canada (NSERC). M. Ghasemi is further sponsored by the Ontario Graduate Scholarship program and the NSERC Canada Graduate Scholarship program. The authors are indebted to the anonymous reviewers for their feedback and suggestions for improvement.

References

1. Abraham J, Reddy MC (2010) Challenges to inter-departmental coordination of patient transfers: a workflow perspective. *Int J Med Informatics* 79(2):112–122. <https://doi.org/10.1016/j.ijmedinf.2009.11.001>
2. Akhigbe O, Amyot D, Anda AA, Lessard L, Xiao D (2016) Consistency analysis for User Requirements Notation models. In: *iStar 2016—ninth international i* workshop*, CEUR-WS, vol 1674, pp 43–48
3. Aldin L, de Cesare S (2011) A literature review on business process modelling: new frontiers of reusability. *Enterprise Inf Syst* 5(3):359–383. <https://doi.org/10.1080/17517575.2011.557443>
4. Amyot D, Ghanavati S, Horkoff J, Mussbacher G, Peyton L, Yu E (2010) Evaluating goal models within the Goal-oriented Requirement Language. *Int J Intell Syst* 25(8):841–877. <https://doi.org/10.1002/int.20433>
5. Amyot D, Horkoff J, Gross D, Mussbacher G (2009) A light-weight GRL profile for i* modeling. In: Heuser CA, Pernul G (eds) *Advances in conceptual modeling—challenging perspectives*. LNCS, vol 5833. Springer, Berlin, Heidelberg, pp 254–364
6. Amyot D, Mussbacher G (2011) User Requirements Notation: the first ten years, the next ten years. *J Softw* 6(5): 747–768
7. Armentano MG, Amandi AA (2012) Towards a goal recognition model for the organizational memory. In: *Computational science and its applications—ICCSA 2012*, LNCS 7335, Springer, pp 730–742 <https://doi.org/10.1007/978-3-642-31137-6>
8. Baek S-J, Ko J-W, Kim G-J, Han J-S, Song Y-J (2012) Goal-heuristic analysis method for an adaptive process mining. In: *Proceedings of the international conference on IT convergence and security 2011*, LNEE 120, pp 409–418. Springer. https://doi.org/10.1007/978-94-007-2911-7_37
9. Ballou RH, Gilbert SM, Mukherjee A (2000) New managerial challenges from supply chain opportunities. *Ind Mark Manage* 29(1):7–18. [https://doi.org/10.1016/S0019-8501\(99\)00107-8](https://doi.org/10.1016/S0019-8501(99)00107-8)
10. Baum LE, Petrie T, Soules G, Weiss N (1970) A maximization technique occurring in the statistical analysis of probabilistic functions of Markov chains. *Ann Math Stat* 41(1):164–171. <https://doi.org/10.1214/aoms/1177697196>
11. Bernard G, Andritsos P (2017) CJM-ex: Goal-oriented exploration of customer journey maps using event logs and data analytics. In: *BPM demo track and BPM dissertation award (BPM-D&DA 2017)*, CEUR-WS Vol-1920 (paper 172)
12. Bernard G, Andritsos P (2017) A process mining based model for customer journey mapping. In: *Proceedings of the forum and doctoral consortium papers presented at the 29th international conference on advanced information systems engineering (CAiSE 2017)*. CEUR-WS vol 1848, pp 46–56

13. Blei DM, Ng AY, Jordan MI (2003) Latent dirichlet allocation. *J Mach Learn Res* 3:993–1022
14. Bresciani P, Perini A, Giorgini P, Giunchiglia F, Mylopoulos J (2004) Tropos: an agent-oriented software development methodology. *Auton Agent Multi-Agent Syst* 8(3):203–236. <https://doi.org/10.1023/B:AGNT.0000018806.20944.ef>
15. Buijs JC, van Dongen B, van der Aalst WMP (2013) Mining configurable process models from collections of event logs. In: *Business process management 2013*, LNCS 8094, pp 33–48, Springer. https://doi.org/10.1007/978-3-642-40176-3_5
16. Cailliau A, van Lamsweerde A (2014) Integrating exception handling in goal models. In: *Requirements engineering conference (RE), 2014 IEEE 22nd International*, pp 43–52. IEEE CS. <https://doi.org/10.1109/re.2014.6912246>
17. Cho M, Song M, Comuzzi M, Yoo S (2017) Evaluating the effect of best practices for business process redesign: an evidence-based approach based on process mining techniques. *Decis Support Syst* 104:92–103. <https://doi.org/10.1016/j.dss.2017.10.004>
18. Chung L, Nixon BA, Yu E, Mylopoulos J (2012) Non-functional requirements in software engineering. In: *International series in software engineering (Vol 5)*. Springer. <https://doi.org/10.1007/978-1-4615-5269-7>
19. Cook JE, Wolf AL (1995) Process discovery and validation through event-data analysis. In: *ICSE'95—Proceedings of the 17th international conference on software engineering*. ACM, pp 73–82
20. Cook JE, Wolf AL (1998) Discovering models of software processes from event-based data. *ACM Trans Softw Eng Methodol* 7(3):215–249. <https://doi.org/10.1145/287000.287001>
21. Dąbrowski J (2017) Towards an adaptive framework for goal-oriented strategic decision-making. In: *Requirements engineering conference (RE), 2017 IEEE 25th international*. IEEE, pp 538–543. <https://doi.org/10.1109/re.2017.53>
22. Dabrowski J, Kifetew FM, Muñante D, Letier E, Siena A, Susi A (2017) Discovering requirements through goal-driven process mining. In: *2017 IEEE 25th international requirements engineering conference workshops REW, IEEE*, pp 199–203. <https://doi.org/10.1109/rew.2017.61>
23. Dees M, de Leoni M, Mannhardt F (2017) Enhancing process models to improve business performance: a methodology and case studies. In: Panetto H, Debruyne C, Gaaloul W, Papazoglou M, Paschke A, Ardagna CA, Meersman R (eds) *OTM confederated international conferences “On the move to meaningful internet systems”*. Springer, Cham, pp 232–251. https://doi.org/10.1007/978-3-319-69462-7_15
24. Deneckère R, Hug C, Khodabandelou G, Salinesi C (2014) Intentional process mining: discovering and modeling the goals behind processes using supervised learning. *Int J Inf Syst Model Design (IJISMD)* 5(4):22–47. <https://doi.org/10.4018/ijismd.2014100102>
25. Dowson M (1987) Iteration in the software process; review of the 3rd international software process workshop. In: *ICSE'87 - proceedings of the 9th international conference on software engineering*, pp 36–41. IEEE CS
26. EDImine (2011) EDImine—mining inter-organizational business processes. Retrieved from <http://edimine.ec.tuwien.ac.at/>
27. El Kharbili M, de Medeiros AKA, Stein S, van der Aalst WM (2008) Business process compliance checking: current state and future challenges. In: *Modellierung betrieblicher Informationssysteme (MobIS 2018)*, LNI 141, pp 107–113. GI-Edition
28. Epure EV, Hug C, Deneckère R, Brinkkemper S (2013) Intention-mining: a solution to process participant support in process aware information systems. Technical Report 2013-020, Department of Information and Computing Sciences, Utrecht University, The Netherlands
29. Epure EV, Hug C, Deneckère R, Brinkkemper S (2014) What shall I do next? Intention mining for flexible process enactment. *Advanced information systems engineering LNCS*, vol 8484. Springer, Cham, pp 473–487. https://doi.org/10.1007/978-3-319-07881-6_32
30. Fahland D, van der Aalst WMP (2015) Model repair—aligning process models to reality. *Inf Syst* 47:220–243. <https://doi.org/10.1016/j.is.2013.12.007>
31. Feldt R, Magazinius A (2010) Validity threats in empirical software engineering research—an initial survey. In: *SEKE 2010—proceedings of the 22nd international conference on software engineering and knowledge engineering*. KSI Research Inc, pp 374–379
32. Flapper SDP, Fortuin L, Stoop PPM (1996) Towards consistent performance management systems. *Int J Oper Prod Manag* 16(7):27–37. <https://doi.org/10.1108/01443579610119144>
33. Fluxicon (2016). Disco. Retrieved from fluxicon.com/disco/
34. Forney GD (1973) The viterbi algorithm. *Proc IEEE* 61(3):268–278. <https://doi.org/10.1109/PROC.1973.9030>
35. Georgeff M, Rao A (1998) Rational software agents: from theory to practice. In: *Agent technology—foundations, applications, and markets*. Springer, pp 139–160. https://doi.org/10.1007/978-3-662-03678-5_8
36. Ghanavati S, Amyot D, Peyton L (2011) A systematic review of goal-oriented requirements management frameworks for business process compliance. In: *2011 fourth international workshop on Requirements engineering and law (RELAW)*, IEEE CS. pp 25–34. <https://doi.org/10.1109/relaw.2011.6050270>
37. Ghasemi M (2018) Towards goal-oriented process mining. In: *Requirements engineering conference (RE), IEEE 26th international*. IEEE CS, pp 484–489. <https://doi.org/10.1109/re.2018.00066>
38. Ghasemi M (2018) What requirements engineering can learn from process mining. In: *1st international workshop on learning from other disciplines for requirements engineering (D4RE)*, IEEE CS, pp 8–11. <https://doi.org/10.1109/d4re.2018.00008>
39. Ghasemi M, Amyot D (2016) Process mining in healthcare: a systematised literature review. *Int J Electron Healthcare* 9(1):60–88. <https://doi.org/10.1504/IJEH.2016.078745>
40. Groen EC, Seyff N, Ali R, Dalpiaz F, Doerr J, Guzman E, Hosseini M, Marco J, Oriol M, Perini A, Stade M (2017) The crowd in requirements engineering: the landscape and challenges. *IEEE Softw* 34(2):44–52. <https://doi.org/10.1109/ms.2017.33>
41. Gurgun Erdogan T, Tarhan A (2018) A goal-driven evaluation method based on process mining for healthcare processes. *Appl Sci* 8(6):894. <https://doi.org/10.3390/app8060894>
42. Hallerbach A, Bauer T, Reichert M (2010) Capturing variability in business process models: the Provop approach. *J Softw Main Evol Res Practice* 22(6–7):519–546. <https://doi.org/10.1002/smr.v22:6/7>
43. Hammer M, Champy J (1993) Reengineering the corporation: a manifesto for business revolution. Zondervan. [https://doi.org/10.1016/s0007-6813\(05\)80064-3](https://doi.org/10.1016/s0007-6813(05)80064-3)
44. Hartigan JA, Wong MA (1979) Algorithm AS 136: a k-means clustering algorithm. *J R Stat Soc Ser C Appl Stat*, 28(1), 100–108. <https://doi.org/10.2307/2346830>
45. Horita H, Hirayama H, Tahara Y, Ohsuga A (2015) Towards goal-oriented conformance checking. In: *27th international conference on software engineering and knowledge engineering (SEKE 2015)*, KSI Research Inc, pp 722–724
46. Horkoff J, Aydemir FB, Cardoso E, Li T, Maté A, Paja E, Salnitri M, Piras L, Mylopoulos J, Giorgini P (2017) Goal-oriented requirements engineering: an extended systematic mapping study. *Require Eng* 1–28. <https://doi.org/10.1007/s00766-017-0280-z>

47. Horkoff J, Barone D, Jiang L, Yu E, Amyot D, Borgida A, Mylopoulos J (2014) Strategic business modeling: representation and reasoning. *Softw Syst Model* 13(3):1015–1041. <https://doi.org/10.1007/s10270-012-0290-8>
48. Horkoff J, Li T, Li F-L, Salnitri M, Cardoso E, Giorgini P, Mylopoulos J (2015) Using goal models downstream: a systematic roadmap and literature review. *Int J Inf Syst Model Design (IJISMD)* 6(2):1–42. <https://doi.org/10.4018/IJISMD.2015040101>
49. Hornix P (2007) Performance analysis of business processes through process mining. Master's thesis, Eindhoven University of Technology, The Netherlands
50. ITU-T (2012) Recommendation Z.151 (10/12): User Requirements Notation (URN)—language definition. <http://www.itu.int/rec/T-REC-Z.151/en>
51. Johann T, Maalej W (2015) Democratic mass participation of users in requirements engineering? In: Requirements engineering conference (RE), 2015 IEEE 23rd international. IEEE CS, pp 256–261 <https://doi.org/10.1109/re.2015.7320433>
52. Khodabandelou G (2013) Contextual recommendations using intention mining on process traces: Doctoral consortium paper. In: IEEE seventh international conference on research challenges in information science (RCIS). IEEE CS, pp 1–6. <https://doi.org/10.1109/rcis.2013.6577728>
53. Khodabandelou G (2014) Mining intentional process models. Doctoral dissertation, University of Paris-Est, France
54. Khodabandelou G, Hug C, Deneckère R, Salinesi C (2013) Process mining versus intention mining. In: Nurcan S, Proper HA, Soffer P, Krogstie J, Schmidt R, Halpin T, Bider I (eds) Enterprise, business-process and information systems modeling. LNBIP, vol 147. Springer, Berlin, Heidelberg, pp 466–480. https://doi.org/10.1007/978-3-642-38484-4_33
55. Khodabandelou G, Hug C, Deneckère R, Salinesi C (2013) Supervised intentional process models discovery using Hidden Markov models. In: IEEE seventh international conference on research challenges in information science (RCIS). IEEE CS, pp 1–11. <https://doi.org/10.1109/rcis.2013.6577711>
56. Khodabandelou G, Hug C, Deneckère R, Salinesi C (2014) Supervised vs. unsupervised learning for intentional process model discovery. In: Bider I, Gaaloul K, Krogstie J, Nurcan S, Proper HA, Schmidt R, Soffer P (eds) Enterprise, business-process and information systems modeling LNBIP, vol 175. Springer, Berlin, Heidelberg, pp 215–229. https://doi.org/10.1007/978-3-662-43745-2_15
57. Khodabandelou G, Hug C, Deneckère R, Salinesi C (2014) Unsupervised discovery of intentional process models from event logs. In: Proceedings of the 11th working conference on mining software repositories—MSR 2014. ACM Press, pp 282–291. <https://doi.org/10.1145/2597073.2597101>
58. Khodabandelou G, Hug C, Salinesi C (2014) A novel approach to process mining: Intentional process models discovery. In: 2014 IEEE eighth international conference on research challenges in information science (RCIS), IEEE CS, pp 1–12 <https://doi.org/10.1109/rcis.2014.6861040>
59. Khodabandelou G, Hug C, Salinesi C (2015) Mining users' intents from logs. *Int J Inf Syst Model Design (IJISMD)* 6(2):43–71. <https://doi.org/10.4018/IJISMD.2015040102>
60. Kingston J, Schafer B, Vandenberghe W (2004) Towards a financial fraud ontology: a legal modelling approach. *Artificial Intell Law* 12(4):419–446. <https://doi.org/10.1007/s10506-005-4163-0>
61. Kitchenham B, Pearl Brereton O, Budgen D, Turner M, Bailey J, Linkman S (2009) Systematic literature reviews in software engineering—a systematic literature review. *Inf Softw Technol* 51(1):7–15. <https://doi.org/10.1016/j.infsof.2008.09.009>
62. Kraiem N, Kaffela H, Dimassi J, Al Khanjari Z (2014) Mapping from MAP models to BPMN processes. *J Softw Eng* 8(4):252–264. <https://doi.org/10.3923/jse.2014.252.264>
63. Krathu W, Engel R, Pichler C, Zapletal M, Werthner H (2013) Identifying inter-organizational key performance indicators from EDIFACT messages. In: 2013 IEEE 15th conference on business informatics, IEEE CS, pp 276–283. <https://doi.org/10.1109/cbi.2013.46>
64. Krathu W, Pichler C, Engel R, Zapletal M, Werthner H, Huemer C (2014) A framework for inter-organizational performance analysis from EDI messages. In: 2014 IEEE 16th conference on business informatics. IEEE CS, vol 1, pp 17–24 <https://doi.org/10.1109/cbi.2014.19>
65. Lapouchnian A, Yu Y, Mylopoulos J (2007) Requirements-driven design and configuration management of business processes. In: Business process management, LNCS 4714, Springer, pp 246–261 https://doi.org/10.1007/978-3-540-75183-0_18
66. Li J, Liu D, Yang B (2007) Process mining: extending α -algorithm to mine duplicate tasks in process logs. In: Advances in web and network technologies, and information management, LNCS 4537, Springer, pp 396–407 https://doi.org/10.1007/978-3-540-72909-9_43
67. Liu L, Yu ESK (2004) Designing information systems in social context: a goal and scenario modelling approach. *Inf Syst* 29(2):187–203. [https://doi.org/10.1016/S0306-4379\(03\)00052-8](https://doi.org/10.1016/S0306-4379(03)00052-8)
68. Ly LT, Maggi FM, Montali M, Rinderle-Ma S, van der Aalst WM (2015) Compliance monitoring in business processes: functionalities, application, and tool-support. *Inf Syst* 54:209–234. <https://doi.org/10.1016/j.is.2015.02.007>
69. Maalej W, Nayebe M, Johann T, Ruhe G (2016) Toward data-driven requirements engineering. *IEEE Softw* 33(1):48–54. <https://doi.org/10.1109/ms.2015.153>
70. Malinova M, Dijkman R, Mendling J (2013) Automatic extraction of process categories from process model collections. In: Business process management workshops, LNBIP 171, Springer, pp 430–441 https://doi.org/10.1007/978-3-319-06257-0_34
71. Maxwell JC, Antón AI, Swire P, Riaz M, McCraw CM (2012) A legal cross-references taxonomy for reasoning about compliance requirements. *Requirements Eng* 17(2):99–115. <https://doi.org/10.1007/s00766-012-0152-5>
72. Mikolov T, Chen K, Corrado G, Dean J (2013) Efficient estimation of word representations in vector space. *arXiv preprint arXiv:1301.3781*
73. Object Management Group (2011) Business process model and notation (BPMN), Version 2.0. Formal/2011-01-03
74. Outmazgin N (2012) Exploring workaround situations in business processes. In: International conference on business process management, Springer, Berlin, pp 426–437 https://doi.org/10.1007/978-3-642-36285-9_45
75. Outmazgin N, Soffer P (2016) A process mining-based analysis of business process work-arounds. *Softw Syst Model* 15(2):309–323. <https://doi.org/10.1007/s10270-014-0420-6>
76. Papadimitriou D, Koutrika G, Mylopoulos J, Velegarakis Y (2016) The goal behind the action: toward goal-aware systems and applications. *ACM Trans Database Syst* 41(4), Article 23. <https://doi.org/10.1145/2934666>
77. Pitman MA (1998) Qualitative research design: an interactive approach. *Anthropol Education Quarterly* 29(4):499–501. <https://doi.org/10.1525/aeq.1998.29.4.499>
78. Poelmans S (1998) Coping strategies and distributed viscosity in a workflow management system: a case study. In: Workshop on adaptive workflow systems, Seattle, USA, November (8 pages)
79. Poelmans S (1999) Workarounds and distributed viscosity in a workflow system: a case study. *ACM SIGGROUP Bulletin* 20(3):11–12. <https://doi.org/10.1145/605610.605618>

80. Poels G, Decreus K, Roelens B, Snoeck M (2013) Investigating goal-oriented requirements engineering for business processes. *J Database Manag* 24(2):35–71. <https://doi.org/10.4018/jdm.2013040103>
81. Ponnalagu K, Ghose A, Narendra NC, Dam HK (2015) Goal-aligned categorization of instance variants in knowledge-intensive processes. In: *Business process management*, LNCS 9253, Springer, pp 350–364 <https://doi.org/10.1007/978-3-319-23063-4>
82. Pourshahid A, Amyot D, Peyton L, Ghanavati S, Chen P, Weiss M, Forster AJ (2009) Business process management with the User Requirements Notation. *Electron Commerce Res* 9(4):269–316. <https://doi.org/10.1007/s10660-009-9039-z>
83. ProM Tools (2016). Retrieved from <http://www.promtools.org/doku.php>
84. Rabiner L, Juang B (1986) An introduction to hidden Markov models. *IEEE ASSP Magazine* 3(1):4–16. <https://doi.org/10.1109/MASSP.1986.1165342>
85. Rabiner LR (1989) A tutorial on hidden Markov models and selected applications in speech recognition. *Proc IEEE* 77(2):257–286. <https://doi.org/10.1109/5.18626>
86. Reijers HA, Mansar SL (2005) Best practices in business process redesign: an overview and qualitative evaluation of successful redesign heuristics. *Omega* 33(4):283–306. <https://doi.org/10.1016/j.omega.2004.04.012>
87. Rolland C, Prakash N, Benjamin A (1999) A multi-model view of process modelling. *Requirements Eng* 4(4):169–187. <https://doi.org/10.1007/s007660050018>
88. Rolland C, Salinesi C (2005) Modeling goals and reasoning with them. *Engineering and managing software requirements*, Springer, Berlin, pp 189–217 https://doi.org/10.1007/3-540-28244-0_9
89. Rozinat A, van der Aalst WMP (2008) Conformance checking of processes based on monitoring real behavior. *Inf Syst* 33(1):64–95. <https://doi.org/10.1016/j.is.2007.07.001>
90. Rubin VA, Mitsyuk AA, Lomazova IA, van der Aalst WM (2014) Process mining can be applied to software too! In: *Proceedings of the 8th ACM/IEEE international symposium on empirical software engineering and measurement*. ACM, p 57. <https://doi.org/10.1145/2652524.2652583>
91. Saiedian H, Kumarakulasingam P, Anan M (2005) Scenario-based requirements analysis techniques for real-time software systems: a comparative evaluation. *Requirements Eng* 10(1):22–33. <https://doi.org/10.1007/s00766-004-0192-6>
92. Santiputri M, Deb N, Khan MA, Ghose A, Dam H, Chaki N (2017) Mining goal refinement patterns: distilling know-how from data. In: *International conference on conceptual modeling*, Springer, Cham, pp 69–76 https://doi.org/10.1007/978-3-319-69904-2_6
93. Snijders R, Dalpiaz F, Hosseini M, Shahri A, Ali R (2014) Crowd-centric requirements engineering. In: *Proceedings of the 2014 IEEE/ACM 7th international conference on utility and cloud computing*. IEEE Computer Society, pp 614–615 <https://doi.org/10.1109/ucc.2014.96>
94. Soffer P (2013) A state-based intention driven declarative process model. *Int J Inf Syst Model Design (IJISMD)* 4(2):44–64. <https://doi.org/10.4018/jismd.2013040103>
95. Spoletini P, Ferrari A (2017) Requirements elicitation: a look at the future through the lenses of the past. In: *Requirements engineering conference (RE)*, 2017 IEEE 25th international, IEEE CS, pp 476–477 <https://doi.org/10.1109/re.2017.35>
96. Sutcliffe A (2003) Scenario-based requirements engineering. In: *11th IEEE international requirements engineering conference*, IEEE CS, pp 320–329 <https://doi.org/10.1109/icre.2003.1232776>
97. Sutcliffe AG, Maiden NA, Minocha S, Manuel D (1998) Supporting scenario-based requirements engineering. *IEEE Trans Softw Eng* 24(12):1072–1088. <https://doi.org/10.1109/32.738340>
98. Taghiabadi ER, Fahland D, van Dongen BF, van der Aalst WM (2013) Diagnostic information for compliance checking of temporal compliance requirements. In: *International conference on advanced information systems engineering*, Springer, Berlin, pp 304–320 https://doi.org/10.1007/978-3-642-38709-8_20
99. Tran H, Zdun U, Dustdar S (2008) Modeling human aspects of business processes—a view-based, model-driven approach. In: Schieferdecker I, Hartman A (eds) *Model driven architecture-foundations and applications*. LNCS, vol 5095. Springer, Berlin, Heidelberg, pp 246–261. https://doi.org/10.1007/978-3-540-69100-6_17
100. van der Aalst WMP (2011) *Process mining: discovery, conformance and enhancement of business processes*. Springer, Berlin. <https://doi.org/10.1007/978-3-642-19345-3>
101. van der Aalst WMP (2012) What makes a good process model? *Softw Syst Model* 11(4):557–569. <https://doi.org/10.1007/s10270-012-0265-9>
102. van der Aalst WMP (2016) *Process mining data science in action* (2nd edn), Springer, Berlin. <https://doi.org/10.1007/978-3-642-19345-3>
103. van der Aalst W, Adriansyah A, van Dongen B (2012) Replaying history on process models for conformance checking and performance analysis. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery* 2(2):182–192. <https://doi.org/10.1002/widm.1045>
104. van der Aalst WMP, Weijters T, Maruster L (2004) Workflow mining: discovering process models from event logs. *IEEE Trans Knowl Data Eng* 16(9):1128–1142. <https://doi.org/10.1109/TKDE.2004.47>
105. van Dongen B, Alves de Medeiros K, Verbeek HMW, Weijters JMM, van der Aalst WMP (2005) The ProM framework: a new era in process mining tool support. In: *Application and theory of petri nets 2005*, LNCS 3536, Springer, pp 444–454 https://doi.org/10.1007/11494744_25
106. van Lamsweerde A (2001) Goal-oriented requirements engineering: a guided tour. In: *RE'01 proc. of the fifth IEEE international symposium on requirements engineering*, IEEE CS, pp 249–261 <https://doi.org/10.1109/isre.2001.948567>
107. van Lamsweerde A (2004) Goal-oriented requirements engineering: a roundtrip from research to practice. In: *Proc 12th IEEE international requirements engineering conference*, 2004, IEEE CS, pp 4–7 <https://doi.org/10.1109/icre.2004.1335648>
108. van Lamsweerde A (2008) Requirements engineering: from craft to discipline. In: *Proceeding SIGSOFT'08/FSE-16*, ACM, pp 238–249 <https://doi.org/10.1145/1453101.1453133>
109. van Lamsweerde A, Darimont R, Letier E (1998) Managing conflicts in goal-driven requirements engineering. *IEEE Trans Softw Eng* 24(11):908–926. <https://doi.org/10.1109/32.730542>
110. van Lamsweerde A, Letier E (2000) Handling obstacles in goal-oriented requirements engineering. *IEEE Trans Softw Eng* 26(10):978–1005. <https://doi.org/10.1109/32.879820>
111. Wang J, Wong RK, Ding J, Guo Q, Wen L (2013) Efficient Selection of Process Mining Algorithms. *IEEE Tran Services Comput* 6(4):484–496. <https://doi.org/10.1109/TSC.2012.20>
112. Weijters AJMM, Ribeiro JJTS (2011) Flexible heuristics miner (FHM). In: *2011 IEEE symposium on computational intelligence and data mining (CIDM)*, IEEE CS, pp 310–317 <https://doi.org/10.1109/cidm.2011.5949453>
113. Xu X, Jin T, Wei Z, Lv C, Wang J (2016) TCPM: topic-based clinical pathway mining. In: *Connected health: applications, systems and engineering technologies (CHASE)*, 2016 IEEE first international conference on, IEEE CS, pp 292–301 <https://doi.org/10.1109/chase.2016.17>

114. Xu X, Jin T, Wei Z, Wang J (2017) Incorporating topic assignment constraint and topic correlation limitation into clinical goal discovering for clinical pathway mining. *J Healthcare Eng*, 2017. <https://doi.org/10.1155/2017/5208072>
115. Yan J, Hu D, Liao SS, Wang H (2014) Mining agents' goals in agent-oriented business processes. *ACM Trans Manag Inf Syst* 5(4):1–22. <https://doi.org/10.1145/2629448>
116. Yu E (1995) Modelling strategic relationships for process reengineering. Doctoral dissertation, University of Toronto, Canada
117. Zeni N, Kiyavitskaya N, Mich L, Cordy JR, Mylopoulos J (2015) GaiusT: supporting the extraction of rights and obligations for regulatory compliance. *Requirements Eng* 20(1):1–22. <https://doi.org/10.1007/s00766-013-0181-8>